

Innovation-Safe Compression of Financial Strategy Libraries: Semantics, Complexity, and Algorithms

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Abstract

A portfolio research team can reduce maintenance burden by retiring strategies, but an entry that adds nothing to today's trading menu may still enable valuable future research. We ask which entries can be removed while preserving both uses. We give a conditional preservation principle: an operating frontier and a declared closure of reusable capabilities preserve the controlled research and operating process under explicit invariance assumptions. With identity closure, minimum-burden safe retention is exactly classical weighted set cover. This representation supplies preprocessing rules, exact methods, and a transferred greedy guarantee. Counterexamples show that preserving current operation can destroy future research value and that heaviest-safe-first deletion can be arbitrarily far from minimum burden. The computational benchmark retains all 3,907 run outcomes and verifies exact-method agreement wherever the required comparisons are available. A retrospective financial study finds large burden-index reductions and additional research options, but no policy replacement in common equities and a loss from its sole ETF replacement. A calibrated synthetic experiment distinguishes option availability from reliable exercise: a small positive bridge value is not recovered, while a prospectively powered value is. These results establish a specified retention principle and bounded experimental findings, without establishing a market premium.

Keywords: financial strategy libraries, innovation-safe compression, weighted set cover, exact algorithms, model governance

1 The financial retention problem

A portfolio research team accumulates strategies that it must test, monitor, document, and maintain. Some are used in portfolios today. Others remain in the library because their signals, filters, allocation rules, or risk controls may be useful in later research. Maintaining every strategy consumes resources; removing a strategy can reduce that burden. The manager’s problem is to decide which strategies can leave the active library without giving up either current operating choices or capabilities needed to build future strategies.

Consider two strategies that offer the same current portfolio opportunity. One also carries a capability needed by a possible new strategy. If the team retains only the other strategy, today’s opportunity is preserved but the new strategy may become unavailable. A performance-based redundancy check cannot detect that loss. Conversely, keeping every strategy that carries a useful capability can be unnecessarily expensive when several entries preserve the same capabilities. The decision therefore concerns the retained library as a whole.

Here, removal means leaving the active maintenance and validation set. Legal records, source code, research history, and mandatory archives remain subject to their own retention rules. A capability is an explicitly declared input to the modeled research process. The model applies when retaining its designated carriers is what makes that capability available; it does not assume that a strategy label contains all of an institution’s knowledge.

We represent the manager’s two preservation targets in simple terms. The *operating frontier* records the best current operating value offered by the source library in each modeled condition. The *generative closure* records the capabilities available after applying the declared rules for combining the retained strategies’ components. Innovation-safe compression finds a least-burden subset of the source library that preserves both targets. The burden can represent a specified validation or governance index; its interpretation must be fixed before looking at subsequent investment outcomes.

The contribution is a conditional preservation principle for a maintained strategy library, an exact optimization representation of that principle, and counterexamples showing why weaker deletion rules can fail. The financial question concerns the availability of reusable inputs to subsequent research. The covering algorithms used to implement the retention decision are established methods. Our financial evidence measures the declared burden and research opportunities separately from subsequent investment performance.

The mathematical argument follows that decision. First, we specify when the two targets summarize the library’s role in a research process. We prove that preserving them then preserves the process’s operating rewards and future choices. An identity-dependent project supplies a counterexample when the assumption fails. Next, for a model in which capabilities are simply the union of retained components, we show that the decision is a weighted covering problem. This connects the financial specification to established optimization methods (Karp 1972; Chvátal 1979). A separate counterexample proves that repeatedly deleting the heaviest redundant entry can leave an arbitrarily expensive library even when every deletion is safe.

The computational study checks exact agreement and the burdens returned by several methods on a registered synthetic design. The financial application then examines portfolio-strategy libraries formed from point-in-time common equities and ETFs. It finds substantial reductions in a validation-burden index while preserving more research options than a budget-matched library that preserves only current operation. The protected policy exercises none of the additional common-equity options; its single ETF replacement loses held-out value. A calibrated synthetic experiment separately shows why an available positive-value option can remain too difficult to recognize at a small signal-to-noise ratio.

These results distinguish preserving an opportunity, deciding to use it, and realizing value from it. They establish a conditional retention principle and bounded computational and financial findings. The financial application is retrospective development evidence, and its burden index is not a measured institutional cost saving. The assumptions and the counterexamples below state precisely where the principle applies.

2 A model of operation and future research

The model records the strategies a manager can operate and the capabilities available for further research. The controller observes a filtered belief, operates from a verified strategy library, and may initiate positive-duration research that generates and verifies a candidate before admission. The theorem does not derive library invariance from finance primitives; it identifies the conclusion that follows when the primitives pass the explicit factorization checks below. The retention decision is made once, before the controlled process is evaluated.

2.1 Hidden state, filtered belief, and operating profiles

Let X be a finite hidden market-state set with at least two elements and let B be a finite nonempty set of filtered beliefs. A belief $b \in B$ has an exact rational interpretation $\mu_b \in \Delta_{\mathbb{Q}}(X)$; the already-filtered belief evolves under an exact rational Markov kernel P_B . The controller observes b , not the hidden state $x \in X$.

The finite strategy catalog is S , with distinguished inactive strategy s_0 . Strategy s has immutable state-contingent operating payoff $u_s : X \rightarrow \mathbb{Q}$ and belief-indexed operating profile

$$j_s(b) := \sum_{x \in X} \mu_b(x) u_s(x), \quad j_{s_0}(b) = 0. \quad (2.1)$$

Thus $j_s(b)$ is the payoff from operating s now; research costs and continuation values are separate objects. The formal projection theorem uses the resulting exact profile table as primitive data, so it does not claim a kernel-verified derivation of that table from a hidden-state model.

Partial information matters to the controlled projection through the filtered belief and its transition law. It is not required by the later tagged-cover equivalence, which needs only a finite index set of operating-profile rows. The financial study in Section 8 uses calendar-year performance profiles crossed with declared cost and risk settings as these rows; it does not estimate the hidden-state filter in this subsection.

2.2 Modules, closure, and the raw retained library

Let M be a finite module set. Each strategy carries an immutable set $\text{mods}(s) \subseteq M$, with $\text{mods}(s_0) = \emptyset$. A module is a declared reusable capability tag for later research, not a direct operating payoff or necessarily a separately owned software artifact. The declared closure operator

$$\text{cl} : 2^M \longrightarrow 2^M$$

is extensive, monotone, and idempotent. These properties allow capabilities to be generated from combinations of raw modules; identity closure is the special case $\text{cl}(A) = A$.

A raw retained library is a source subset $L \subseteq S$ containing s_0 . Its operating frontier, raw module inventory, and generative closure are

$$F_L(b) := \max_{s \in L} j_s(b), \quad U_L^{\text{raw}} := \bigcup_{s \in L} \text{mods}(s), \quad (2.2)$$

$$C_L := \text{cl}(U_L^{\text{raw}}). \quad (2.3)$$

Finiteness and mandatory inactivity make the maximum attained and imply $F_L(b) \geq 0$. The raw decision state is (b, L) : continuing earns the frontier payoff, advances the belief once, and leaves the library unchanged.

An entry may itself be a portfolio strategy, but the operating decision here selects one catalog entry at a time. Equality of F_L therefore preserves the best entry's modeled payoff at each belief; it does not by itself preserve the risk–return opportunities from combining entries with different joint returns. An application that reallocates across entries must include the relevant joint payoff and allocation information in its preservation target.

2.3 Generation, verification, admission, and timing

Let Q be a finite project set. At project q , belief b , and closure C , candidate generation follows an exact law

$$G(q, b, C) \in \Delta_{\mathbb{Q}}(\{\text{none}\} \dot{\cup} S).$$

A generated candidate s passes verification with exact probability $\nu(q, b, C, s)$. The admitted law is derived rather than separately chosen:

$$\Gamma(q, b, C)(s) = G(q, b, C)(s)\nu(q, b, C, s), \quad (2.4)$$

and all failed-generation and failed-verification mass is assigned to **none**. Successful admission adds the verified catalog row, whereas every other outcome leaves the source unchanged:

$$L \oplus \text{none} = L, \quad L \oplus s = L \cup \{s\}. \quad (2.5)$$

Each available project has an initiation cost, an operation/suspension flag, and an integer duration $d_q \geq 1$. Its exact joint completion law couples the intervening belief path with the admitted outcome; the two components need not be independent. The incumbent library remains in force until completion, then (2.5) is applied. Full

probability-table definitions, null-outcome normalization, path and outcome marginals, and pushforward identities are assigned to Online Resource 1, Section S1.

2.4 The productive library state

Definition 2.1 (Frontier-closure library state). The productive state of a retained library is

$$K_L := (F_L, C_L). \quad (2.6)$$

It is the *library component* of the controlled state. The filtered belief remains part of the decision state, and an unfinished project also requires its identity and remaining duration.

For $K = (F, C)$, define the local compressed update

$$\text{addK}(K, \text{none}) = K, \quad \text{addK}(K, s) = (b \mapsto \max\{F(b), j_s(b)\}, \text{cl}(C \cup \text{mods}(s))).$$

Extensivity and idempotence of closure give

$$K_{L \oplus o} = \text{addK}(K_L, o). \quad (2.7)$$

Assumption 2.2 (Finite library-invariant research process). The sets B, S, M, Q are finite, B and every admissible library are nonempty, and all displayed profiles and probability tables are exact. The inactive strategy has zero profile and no modules; closure is extensive, monotone, and idempotent. Generation and primitive verification depend on the incumbent library only through (q, b, C_L) . Project availability, initiation cost, duration, operation flag, and the joint completion law depend on the library only through (b, K_L) . Each duration is positive, the joint law has the declared belief-path and admitted-outcome marginals, and discount satisfies $0 \leq \beta < 1$. Stationary conclusions additionally assume the recorded raw and compressed Bellman operators are contractions.

3 What safe retention preserves

Theorem 3.1 (Raw-to-compressed controlled Markov projection). *Under Assumption 2.2, the update in (2.7) induces a normalized transition on realizable states K_L . If $K_L = K_{\tilde{L}}$, the two raw libraries have identical project availability, cost, duration, incumbent operating reward blocks, and joint terminal-belief-next-compressed-state laws after projection. Hence (b, K_L) is a controlled semi-Markov projection at embedded decision epochs and, for every finite calendar horizon h ,*

$$V_h^{\text{raw}}(b, L) = \bar{V}_h(b, K_L).$$

Under the additional contraction clause, stationary fixed-point values agree under the same projection and a maximizing stationary compressed selector lifts to a Bellman-optimal raw selector.

Proof sketch. Equation (2.7) makes the next compressed state a deterministic pushforward of the admitted raw update. The library-invariance assumptions make this law, every feasible action, and its payoff block constant on each fiber of $L \mapsto K_L$. Strong induction on calendar horizon is well founded because every research duration is positive; corresponding raw and compressed action values, and then their finite maxima, agree. Bellman intertwining and contraction give the stationary clauses. The complete normalization and projection proof is in Online Resource 1, Sections S1–S2. The result permits belief–admission dependence and makes no generic quotient-minimality claim.

Operational factorization audit.

The assumption can be checked table by table. Project availability, cost, duration, operation while active, and the joint completion law must be constant whenever (b, F_L, C_L) is constant; candidate generation and verification must be constant whenever (q, b, C_L) is constant; and admission must update the compressed state through addK. A nontrivial finite class satisfying these conditions consists of module-gated projects whose availability is a predicate of (q, b, C_L) , whose candidate and verification tables are indexed by (q, b, C_L) , and whose duration, cost, and completion tables are indexed by (q, b, F_L, C_L) . Correlation between terminal belief and admission remains unrestricted within each indexed joint table.

The condition is substantive. Let two one-active-strategy libraries $L_a = \{s_0, a\}$ and $L_b = \{s_0, b\}$ have identical zero profiles and no modules. Then $K_{L_a} = K_{L_b}$. If a zero-cost, one-period project with deterministic positive completion value is available only when the raw identity a is retained, their feasible actions and values differ. The projection therefore fails. The same failure occurs if a primitive depends directly on raw cardinality, provenance, owner, or archival status. These variables must either be included in the compressed state or ruled out by the declared information architecture.

3.1 Retention burden and exact source-relative compression

Let $w_{s_0} = 0$ and give every active strategy an exact rational retention weight $w_s > 0$. The one-time burden is

$$W(L) := \sum_{s \in L} w_s. \quad (3.1)$$

It may index storage, validation, monitoring, active maintenance, retrieval, governance, and human attention. Additivity is a modeling assumption. A weight is not automatically an observed monetary or labor cost, and it is not operating payoff or project cost. Burden is not a component of K_L : equal productive states can have different burdens. In the financial cases, “retained” means retained in the active maintenance and validation set; mandatory archival records, code, data, ownership, and reactivation controls remain outside this binary decision.

For a fixed verified source L , exact source-relative compression chooses an inactive-containing sublibrary $L' \subseteq L$ with $K_{L'} = K_L$ and minimum $W(L')$. By Theorem 3.1, this outer retention decision changes only the raw representation, not the declared productive process. Section 4 defines its feasible family and optimization model.

4 Exact compression, covering representation, and complexity

Fix a verified finite source library L . Safety is source-relative: compression may delete source entries but may neither import a replacement nor change the declared profiles, modules, or closure operator.

4.1 Safe feasibility and productive preservation

The safe-feasible family and its global minimum-weight correspondence are

$$\mathcal{P}_{\text{safe}}(L) := \{L' \subseteq L : s_0 \in L', F_{L'} = F_L, C_{L'} = C_L\}, \quad (4.1)$$

$$\mathcal{M}_{\text{safe}}(L) := \arg \min_{L' \in \mathcal{P}_{\text{safe}}(L)} W(L'). \quad (4.2)$$

Equality is exact at every declared belief and for the entire declared generative closure. In particular, a currently dominated strategy is not safe to remove merely because it contributes no current frontier value.

Theorem 4.1 (Minimum-resource innovation-safe representation). *Let L be a finite source library containing s_0 , let the catalog and closure satisfy Assumption 2.2, and let $w_{s_0} = 0$ with exact rational $w_s > 0$ for every active source strategy. Then $\mathcal{P}_{\text{safe}}(L)$ is finite and nonempty, $\mathcal{M}_{\text{safe}}(L) \neq \emptyset$, and every $L' \in \mathcal{P}_{\text{safe}}(L)$ has the same finite-horizon productive values as L . Under the contraction clause of Assumption 2.2, it also has the same stationary productive value and complete optimal stationary-action correspondence.*

Proof. The source is feasible relative to itself and has only finitely many sublibraries, so an exact burden minimum is attained. Feasibility is exactly $K_{L'} = K_L$; the value and action conclusions follow from Theorem 3.1. Neither optimizer uniqueness nor a particular deletion path follows.

A converse from equal observed behavior to equal closure requires an additional detectability assumption; the supporting proof states that separate condition.

4.2 Identity closure induces a tagged cover

The reduction in this subsection is the computational consequence of the preservation semantics. Once the tagged universe is constructed, the resulting optimization problem is ordinary weighted set cover: its generic binary model, NP-hardness antecedent, greedy approximation theorem, mask recurrence, and standard cover preprocessing are not claimed as new combinatorial-optimization theory. The application-specific content is the exact mapping from frontier and closure equality to tagged coverage, together with its identity-closure limit.

The algorithms below apply to identity closure. General closure remains governed by exact equality of the closed module unions.

Assume throughout this subsection that closure is identity. Define the disjoint tagged requirement universe

$$\mathcal{U}_L^{\text{tag}} := \{\text{frontier}(b) : b \in B\} \dot{\cup} \{\text{module}(m) : m \in C_L\}, \quad (4.3)$$

and, for each $s \in L$, its source-relative coverage

$$\mathcal{R}_s := \{\text{frontier}(b) : j_s(b) = F_L(b)\} \cup \{\text{module}(m) : m \in \text{mods}(s)\}. \quad (4.4)$$

Frontier and module tags remain distinct even when their underlying identifiers print alike. Exact equality in the first term means that every tied source attainer covers the belief row.

Theorem 4.2 (Tagged-cover equivalence under identity closure). *Let B, M, S be finite, let $B \neq \emptyset$, and let $s_0 \in L' \subseteq L \subseteq S$. Profiles are exact rational functions $j_s : B \rightarrow \mathbb{Q}$; $j_{s_0} = 0$, $\text{mods}(s_0) = \emptyset$, maxima are attained, and closure is identity on 2^M . Then*

$$F_{L'} = F_L \text{ and } C_{L'} = C_L \iff \mathcal{U}_L^{\text{tag}} \subseteq \bigcup_{s \in L'} \mathcal{R}_s. \quad (4.5)$$

No unique frontier attainer or unique module carrier is assumed. If $F_L(b) = 0$, mandatory s_0 itself covers $\text{frontier}(b)$.

Proof. If L' preserves the frontier, a finite maximizer in L' covers each frontier tag. Under identity closure, $C_{L'} = C_L$ is equality of raw module unions, so some retained carrier covers every module tag. Conversely, source inclusion gives $F_{L'} \leq F_L$ and $C_{L'} \subseteq C_L$. Coverage supplies a retained source attainer at every belief and a retained raw carrier for every source module, giving the reverse inclusions. Ties and multiple carriers are existential choices, while the zero-frontier row is witnessed by s_0 .

For exact rational weights, the equivalent binary model is

$$\min_x \sum_{s \in L} w_s x_s, \quad (4.6)$$

$$\text{s.t.} \quad \sum_{s \in L: j_s(b) = F_L(b)} x_s \geq 1, \quad b \in B, \quad (4.7)$$

$$\sum_{s \in L: m \in \text{mods}(s)} x_s \geq 1, \quad m \in C_L, \quad (4.8)$$

$$x_{s_0} = 1, \quad x_s \in \{0, 1\}, \quad s \in L. \quad (4.9)$$

The source restriction is built into the columns. All zero-frontier rows may be removed after fixing s_0 , but retaining them in $\mathcal{U}_L^{\text{tag}}$ keeps the semantic construction literal.

The boundary is substantive. If $\text{cl}(\{a, b\}) = \{a, b, c\}$ although no strategy carries c , an independent row for each member of C_L rejects even the source library. General closure therefore requires direct closure equality or a separately proved extended

formulation. Frontier rows alone and module rows alone are likewise insufficient; exact counterexamples are in Online Resource 1, Section S3.

4.3 Baseline mixed-integer formulation

Equations (4.6)–(4.9) are the baseline MIP. Its input is a validated, canonically ordered identity-closure instance with exact weights and declared mandatory columns. Exact preprocessing may fix columns and remove satisfied rows before JuMP and HiGHS receive the residual binary cover. Exact weights are scaled to an integer objective only after the scaling identity and solver range are verified.

Each returned selection is lifted to original strategy identities and independently checked for mandatory retention, frontier equality, closure equality, and exact burden. Fractional vectors are rejected without rounding. Solver status and bounds remain numerical evidence of global optimality; the exact postcheck certifies the returned library’s feasibility and burden. If preprocessing solves the instance completely, no solver is invoked.

4.4 Computational complexity

IDENTITY-CLOSURE SAFE COMPRESSION (ISC-ID) has the following explicit decision encoding. The input lists a finite source $L = \{s_0, s_1, \dots, s_n\}$, a finite nonempty belief set, exact binary-rational profiles, a finite module universe and incidence rows, identity closure, positive binary-integer active weights, and a nonnegative binary-integer threshold D . Strategy s_0 is mandatory, has zero profile, no modules, and zero weight. The question is whether some $s_0 \in L' \subseteq L$ preserves the exact source frontier and closure and has $W(L') \leq D$.

Theorem 4.3 (Identity-closure safe-compression complexity). *Under the explicit encoding in the preceding paragraph, ISC-ID is NP-complete. It remains NP-complete on the restricted subclass with one belief, zero profiles for every strategy, a zero-weight module-free inactive strategy, identity closure, a source whose active module rows have full union, and unit active weights. Consequently, minimum-burden identity-closure safe compression is NP-hard, and any broader model class containing this subclass is NP-hard. No NP-membership conclusion for arbitrary nonidentity closure follows unless its encoding admits a polynomial exact closure-equality verifier.*

Proof sketch. A selected-strategy bit vector is a polynomial certificate: exact rational maxima, row unions, binary-integer burden, source inclusion, and mandatory retention are polynomially checkable in the explicit input length. For hardness, normalize Karp’s SET COVER decision problem (Karp 1972) to a nonempty universe and a full-union indexed family, using fixed yes/no outputs for the empty- or uncovered-universe branches. Give each universe element a module, each indexed set an active strategy carrying its modules, every active strategy unit weight and zero profile, and add one zero-profile inactive strategy. Use one belief and copy the budget. The inactive strategy preserves the trivial frontier, while identity-closure equality is exactly set-cover equality and burden is exactly selected-set cardinality. Incidence size and integer bit lengths

are copied polynomially. The complete membership, normalization, forward, reverse, size, and threshold audit is in Online Resource 1, Section S3. No strong-hardness, inapproximability, or APX-hardness claim is made.

4.5 Exact preprocessing of the tagged cover

Fix the tagged universe from the source before reducing the model. After declared mandatory strategies are selected, let Q be the unsatisfied rows, A the available nonfixed columns, and $\mathcal{R}_s^Q = \mathcal{R}_s \cap Q$. Mandatory weights are nonnegative, optional weights are strictly positive exact rationals, and there are no identity-sensitive side constraints, groups, exact-cardinality conditions, or mutual exclusions beyond the pure cover. All incidence and weight comparisons are exact.

Proposition 4.4 (Exact preprocessing suite). *Under the assumptions in the preceding paragraph, the following fixed-point rules preserve feasibility, the exact optimum value including forced-weight offsets, and at least one liftable optimum:*

1. fix the unique available carrier of any residual requirement;
2. delete nonmandatory s when $\mathcal{R}_s^Q \subset \mathcal{R}_t^Q$ and $w_t \leq w_s$;
3. for identical residual coverage, retain a stable minimum-weight representative and record every equal-weight substitution;
4. after a forced selection, remove satisfied rows and recompute all rules;
5. delete a nonmandatory column with empty residual coverage; and
6. merge rows having identical carrier sets.

Strictly cheaper dominance, strictly heavier duplicates, empty positive-weight columns, forced columns, and duplicate rows preserve all original optimizer identities. Equal-weight duplicate columns recover all identities through the recorded substitution map. Equal-weight strict dominance need not preserve all optimizer identities and makes no such claim. Restarting the stable scan after each transformation terminates after finitely many row or column removals and yields a fixed point. A zero-carrier residual row is an infeasibility certificate, not an invitation to repair the instance.

Proof sketch. A unique carrier is forced by its covering row. A no-heavier coverage superset can replace a dominated selected column without losing a witness. Identical columns permit exact substitution; identical rows impose the same inequality; and an empty positive-weight column occurs in no optimum. Forced columns contribute a selection-independent offset, and propagation composes these correspondences. Every successful step removes a finite row or column, which proves termination. The strict/equal weight cases give precisely the optimizer-identity qualifications in the statement. Full propositions, adversarial examples, reconstruction maps, and the deterministic rule order are in Online Resource 1, Section S4.

5 Algorithms and guarantees

All methods in this section consume the same validated identity-closure instance. Strategies and tagged requirements have canonical original order; weights are exact

rational; mandatory flags, source profiles, raw module memberships, the source frontier, and the source closure are stored independently of the incidence matrix. Preprocessing supplies an exact objective offset and a reconstruction map. The number of residual optional strategies is n , the number of residual tagged requirements is r , and strategy i has residual mask R_i and positive weight w_i .

5.1 Common output and certificate

Every algorithm returns original strategy identifiers in canonical order, the exact burden, its declared completion status, its tie rule, an algorithm trace or search counters, and any required random seed. A result is reportable only after an independent certificate recomputes from the original catalog:

1. every mandatory strategy is retained;
2. every tagged requirement is covered;
3. the complete source operating frontier is preserved exactly;
4. the complete source identity closure is preserved exactly; and
5. the exact burden agrees with the returned burden.

Passing this certificate establishes feasibility and burden for the returned library. It does not establish heuristic optimality, convert a solver status into a proof, or show that two implementations are independent. Complete-tie output is permitted only when the preprocessing map declares every optimizer identity reconstructable; otherwise the result is one explicitly declared representative.

Table 1 Guarantees under the stated assumptions. Exact returned-library checks certify feasibility; they do not turn solver or heuristic output into a proof of optimality.

Method	Guarantee and limit
Enumeration	Exact optimum; exponential in optional strategies n .
Mask dynamic programming	Exact optimum; $O(n2^r)$ recurrence operations, exponential in residual requirements r .
HiGHS mixed-integer model	Exactly rechecked feasible incumbent; solver-reported optimality is separate evidence.
Weighted greedy, with optional reverse deletion	$H(d) \leq H(r)$ approximation under identity closure and positive optional weights.
Cardinality greedy	Feasible cover; no burden factor for unequal weights.
Rechecked deletion, including multi-start	Feasible, one-deletion irreducible endpoint; no general burden factor. Heaviest-first has an unbounded worst-case ratio.

5.2 Independent exact methods

5.2.1 Complete enumeration

Proposition 5.1 (Correctness of complete enumeration). *Let the input be a valid finite identity-closure tagged instance with exact nonnegative mandatory weights, strictly*

positive optional weights, at least one carrier for every residual requirement, and a valid preprocessing reconstruction map. Complete enumeration returns a minimum-burden safe source sublibrary after checking all 2^n residual selections. If complete ties are requested, the map preserves all optimizer identities, and the output cap is not exceeded, it returns every original optimizer; otherwise it returns the declared stable representative.

Proof. Every admissible library is a forced set plus one residual subset. Enumeration evaluates every such subset and takes the exact minimum; the fixed offset and valid lift preserve burden and feasibility. The final certificate maps tagged feasibility back to the original source semantics through Theorem 4.2. Complete reconstruction follows only under the stated map and output-cap conditions. Time is $2^n \text{poly}(|I|)$, so enumeration is exact but exponential in the number of optional strategies.

5.2.2 Requirement-mask dynamic programming

Encode a residual requirement subset by an r -bit mask. With $D_i(q)$ the least exact burden that attains mask q using the first i strategies, the recurrence is

$$D_i(q) = \min \left\{ D_{i-1}(q), \min_{p: p \text{ OR } R_i = q} (D_{i-1}(p) + w_i) \right\}. \quad (5.1)$$

Theorem 5.2 (Exact requirement-mask dynamic programming). *Let the input satisfy every assumption of Proposition 5.1. The recurrence in (5.1) reconstructs at least one minimum-burden safe source sublibrary. Representative mode uses $O(n2^r)$ recurrence operations, $O(2^r)$ exact cost states, and $O(n2^r)$ predecessor records when full reconstruction is retained. Value-only two-layer storage uses $O(2^r)$ memory. Materializing T tied optimizers adds $O(Tn)$ output bits and may be exponential in n . Its bit time is $2^r \text{poly}(|I|)$: the method is fixed-parameter tractable in r , but it is not polynomial in unrestricted input size.*

Proof sketch. Induct on i . At $i = 0$, only the empty subset attains mask zero. Every subset of the first i strategies either skips i , or removes i to give a predecessor mask and then takes it. The two transitions are exhaustive and construct valid subsets, so the recurrence stores the least exact burden for each mask. The all-ones mask is precisely residual feasibility. Offset, lifting, and Theorem 4.2 yield a safe global optimum. The state, transition, memory, and output counts give the displayed bounds. The complete induction and reconstruction proof is in Online Resource 1, Section S5.

5.3 Weighted greedy construction

After selecting every mandatory or forced strategy, let U^0 be the remaining tagged requirements and let

$$d := \max_i |R_i|, \quad H(d) := \sum_{j=1}^d \frac{1}{j}.$$

Theorem 5.3 (Transferred weighted-greedy guarantee). *Let the source be a finite identity-closure instance satisfying Theorem 4.2; let every residual requirement have a carrier; let mandatory weights be nonnegative and every optional weight be positive; and apply only mandatory/forced preprocessing with a valid exact offset and lift. If $U^0 \neq \emptyset$, weighted greedy returns a safe library of burden W_G satisfying*

$$W_G \leq H(d) \text{OPT} \leq H(r) \text{OPT}.$$

Reverse deletion returns a safe, inclusion-wise irreducible library with $W_{GR} \leq W_G$, so it satisfies the same bound. If $U^0 = \emptyset$, the forced library is optimal and the factor is one. The guarantee does not extend to arbitrary nonidentity closure.

Proof sketch. On the residual tagged universe, the rule minimizes weight per newly covered requirement. It is precisely the positive-cost weighted set-cover greedy algorithm of Chvátal (1979), which gives $G \leq H(d) \text{OPT}_R$. Adding a nonnegative fixed burden W_0 preserves the displayed factor. Reverse deletion accepts only feasibility-preserving removals of positive-weight strategies and therefore cannot increase burden. Later removals cannot restore coverage lost by an earlier rejected removal, so one reverse pass is irreducible. The theorem transfers the classical guarantee through Theorem 4.2; it does not reprove the classical set-cover theorem. The cardinality variant replaces w_i by one in the score and has no retention-burden approximation guarantee when weights differ.

With straightforward incidence scans, greedy and reverse deletion each use a polynomial number of exact operations (at most n selection/deletion rounds and $O(nr)$ incidence work per round). This implementation bound does not alter the approximation factor.

5.4 Certified rechecked deletion

Rechecked deletion starts from the full source and never infers safety from a stale certificate. At every scan, each nonmandatory trial deletion is checked against the original source frontier and closure; after an accepted deletion, all candidate certificates are discarded and the scan restarts.

Deletion orders include heaviest-first, lightest-first, minimum unique-carrier exposure, source order, and a seeded random order. Maximum immediate additive release is the same rule as heaviest-first. The score chooses only among freshly certified safe deletions; the complete definitions are in Online Resource 1, Section S5.

Proposition 5.4 (Certified deletion endpoint). *Let the source be finite and safe relative to itself, let all mandatory strategies be protected, and suppose every accepted trial deletion is recomputed exactly and shown to preserve the original frontier, declared closure, and burden identity. Then every deterministic or seeded-random rechecked-deletion variant above terminates at a safe source sublibrary. A complete exact final scan certifies that no retained nonmandatory strategy can be safely deleted. This is a one-deletion irreducibility certificate, not a global-optimality or approximation certificate.*

Proof. The source is safe. Exact certification of each accepted trial preserves safety inductively, and each step removes one strategy, so the trace is finite. The final scan checks every remaining eligible deletion. The score and random order choose only among already certified candidates and therefore do not change the argument. If closure is nonidentity, this proposition requires the separate exact closure checker stipulated in its assumptions.

Multi-start deletion takes the lowest-burden certified endpoint over a fixed, registered number of random orders. Extra starts cannot worsen the best observed burden, but supply no approximation factor.

5.5 Worst-case local–global separation

Theorem 5.5 (Unbounded gap for heaviest-safe-first deletion). *Let $k \geq 2$ be an integer and let $\epsilon \in \mathbb{Q}$ satisfy $0 < \epsilon < k - 1$. Use one belief, identity closure, modules $m_1, \dots, m_k$, and a mandatory inactive strategy with zero profile, zero burden, and no modules. For each i , let singleton s_i have zero profile, weight one, and module set $\{m_i\}$. Let bundle s_B have zero profile, weight $1 + \epsilon$, and all k modules. Rechecked heaviest-safe-first deletes the uniquely heaviest safe bundle, terminates at the k -singleton library of burden k , and returns an inclusion-wise irreducible endpoint. The unique global optimum retains only the bundle and has burden $1 + \epsilon$. The exact ratio is*

$$\frac{k}{1 + \epsilon},$$

so heaviest-safe-first and one-deletion irreducibility alone have no constant burden guarantee over this family. The conclusion does not apply to every deletion order or every certified deletion heuristic.

Proof. All profiles are zero, so safety is exactly coverage of all modules. At the source, the bundle makes every singleton deletable and all singletons together make the bundle deletable. Since $\epsilon > 0$, the bundle is uniquely heaviest and is deleted first; afterward each singleton uniquely carries its module. Any safe library without the bundle must therefore retain all singletons and costs k , whereas the bundle alone is safe and costs $1 + \epsilon < k$; positive weights make it the unique optimum. Fixing, for example, $\epsilon = 1/2$ makes the ratio unbounded as k grows. A singleton-first order instead reaches the bundle optimum, proving the stated order boundary. The complete family proof and finite fixture are in Online Resource 1, Section S6.

5.6 Guarantee and evidence boundary

Enumeration and mask DP are exact finite methods under their stated identity-closure assumptions; the former is exponential in n , and the latter is exponential in r . Weighted greedy is approximate but has the transferred $H(d)$ bound; reverse deletion preserves that bound. Rechecked deletion and multi-start return exactly certified feasible, irreducible endpoints but remain heuristics. The MIP in Section 4.3 supplies solver evidence plus an exact returned-library postcheck. Complete proofs and

machine-readable certificate schemas are in Online Resource 1; benchmark runtimes and method comparisons are experimental evidence reported separately in Section 7.

The negative guarantee concerns this named rule and one-deletion neighborhoods; it makes no claim about stronger exchange search.

6 Why current redundancy can destroy future value

The preservation constraint has an economic purpose beyond reproducing a static payoff menu. Frontier equality protects current operating opportunities; closure equality protects the inputs from which future opportunities can be generated. We separate current operating value from the value of access to a future project.

6.1 Operational and generative value

For a finite horizon, let $P_n(b, L)$ be the value when the library is frozen and research is unavailable, and let $U_n(b, L)$ be the corresponding full value when every feasible positive-duration project is available. Thus

$$\begin{aligned} P_0(b, L) &= 0, \\ P_{n+1}(b, L) &= F_L(b) + \beta \sum_{b'} P_B(b, b') P_n(b', L), \end{aligned} \quad (6.1)$$

where P_B is the filtered-belief transition under continued operation. Write $\Omega_n = U_n - P_n$ for the research-option premium. For inserting a candidate s , define total, operational, and generative increments by

$$\begin{aligned} \mathcal{I}_n(s \mid b, L) &:= U_n(b, L \cup \{s\}) - U_n(b, L), \\ \Delta_n^{\text{op}}(s \mid b, L) &:= P_n(b, L \cup \{s\}) - P_n(b, L), \\ \Delta_n^{\text{gen}}(s \mid b, L) &:= \Omega_n(b, L \cup \{s\}) - \Omega_n(b, L). \end{aligned} \quad (6.2)$$

Proposition 6.1 (Operational–generative decomposition). *For every finite horizon n , filtered belief b , finite raw library L , and admissible candidate s ,*

$$\mathcal{I}_n(s \mid b, L) = \Delta_n^{\text{op}}(s \mid b, L) + \Delta_n^{\text{gen}}(s \mid b, L). \quad (6.3)$$

If $F_{L \cup \{s\}} = F_L$, then the operational increment is zero. If in addition $C_{L \cup \{s\}} = C_L$, then the total increment is zero under the projection assumptions of Theorem 3.1.

The equality is an accounting identity, not a sign theorem. A strategy may be silent in the passive operating problem and valuable because it carries a module that enables research; conversely, the generative residual need not be positive if insertion changes costs or the relative appeal of projects. The frontier–closure projection makes the two channels operationally useful: frontier silence identifies zero current operating value, whereas joint frontier and closure silence identifies zero productive value within the declared process.

6.2 Sharp loss from frontier-only retention

Consider a dominated bridge strategy that uniquely carries the capability needed for one descendant project. Its deletion leaves the source frontier unchanged but makes the distinguished descendant unreachable. The following statement fixes the scope of the loss calculation: the incumbent operating block is common, there are no other differential projects or continuation gains, and the terminal continuation gain is bounded by C .

Theorem 6.2 (Sharp normalized frontier-only bridge loss). *Fix exact rationals $0 \leq \beta < 1$, $0 \leq \rho, \pi \leq 1$, a reward cap $C \geq 0$, a project cost $\kappa \geq 0$, and an integer delay $d \geq 1$. Suppose the bridge uniquely enables the descendant, retention gives survival mass ρ^d and conditional admission probability π , deletion makes that descendant mass zero, and $\kappa \leq \beta^d \rho^d \pi C$. Then the retained-minus-deleted value loss is*

$$\text{Loss}_d = \beta^d \rho^d \pi C - \kappa. \quad (6.4)$$

For fixed $(\beta, \rho, \pi, d, \kappa)$, this is the sharp upper bound over descendant rewards in $[0, C]$, attained at C . Whenever the denominator is positive, the loss divided by the attainable net bridge value equals one. If $C \leq 1$, the additive loss is at most one; arbitrarily large additive loss requires reward scaling.

The proof expands the one available project branch. Retention contributes the discounted joint survival-and-admission mass times the capped continuation gain, less cost; deletion contributes zero. Maximization over the reward interval is attained at its upper endpoint. The factor $\rho^d \pi$ comes from sequential generation and conditional admission, not from an independence assumption. Common operating value cancels under frontier equality; if operation differs while research is in progress, its exact discounted difference must be added. The calculation above supplies the bridge proof; the positive-duration model accounts separately for any operating reward earned during research.

The result explains why operational redundancy is not generative redundancy. A frontier-only rule can be exact for today's operating menu and still remove the entire positive net value of a future bridge. It does not assert an unbounded normalized loss or a general value formula for arbitrary deleted libraries.

7 Registered computational study

The registered algorithmic compression benchmark is a finite synthetic study of tagged identity-closure instances. It can test exact-method agreement, certified burden quality, preprocessing prevalence, and the declared adversarial mechanisms. It cannot establish average-case complexity, steady-state runtime rankings, or a population law for financial libraries.

7.1 Registered design, execution, and audit

The final registry contains 173 analysis instances: 24 small exact, 128 structured, and 21 adversarial. Family B requested variation in strategy and tagged-row counts, coverage density, module overlap, bundle prevalence, unique-carrier frequency, weight dispersion, and frontier-module association. Family C contains the predeclared gap, tie, duplicate, dominance, rare-carrier, symmetry, and bundle mechanisms. Failed generation remains a terminal result; no failed row is silently repaired or dropped.

The algorithm grid contains enumeration, requirement-mask DP, two registered HiGHS formulations, weighted and cardinality greedy, greedy plus reverse deletion, rechecked deletion orders, seeded random deletion, and 32-start deletion. Maximum immediate additive burden release is exactly the heaviest-safe-first ordering. The two labels denote one method in all scientific comparisons.

The audit passed all 3907 registered run units and accepted 3526 returned libraries after original-coordinate mandatory, coverage, frontier, identity-closure, and exact-burden checks. It retained 381 unsuccessful records in their terminal categories. HiGHS OPTIMAL remains solver evidence plus an exact incumbent check, not formal or exhaustive proof.

7.2 Exact agreement and applicability

Among 38 registry instances, all required exact-method burdens were available and equal on 31; 7 did not have a complete comparison. 6 adversarial enumeration jobs are recorded as `IMPLEMENTATION_ERROR`, but their substantive cause is known: the residual optional count exceeded the declared enumeration cap. The recorded status is retained in the failure census. They are enumeration-inapplicable under that cap, and exact agreement is unavailable rather than inferred. Optimizer identities differed at equal burden on 9 complete comparisons, which records valid tied optima rather than disagreement.

These are exact finite facts about specified instances. The medium structured references are HiGHS-optimal incumbents that pass exact feasibility and burden rechecks; no second solver was pinned.

7.3 Certified burden quality

Table 2 separates Family A exact finite references from Family B exactly rechecked HiGHS-optimal references. Within those distinct evidence classes, weighted greedy plus reverse deletion and heaviest-safe-first often attain the available reference, while several other deletion orders have larger observed tails. Multi-start attains every available Family A reference but not every Family B reference. These are descriptive outcomes of the registered instances, not approximation guarantees or an algorithm ranking beyond the design.

The registered heaviest-safe-first gap family gives a different kind of evidence. Its exactly rechecked finite ratios match Theorem 5.5 at every registered scale. The theorem establishes the universal family statement; the experiment checks the implementation on the named fixtures.

The supporting benchmark appendix reports the finite ratios.

Algorithm	Attained/reference	Mean gap	Median	q90	Worst
<i>Family A: exact finite reference</i>					
Weighted greedy	16/23	0.08684	0	0.25	0.5
Cardinality greedy	13/23	0.6375	0	1.4	7
Weighted greedy + reverse deletion	20/23	0.02923	0	0.16	0.25
Heaviest safe first	22/23	0.008696	0	0	0.2
Lightest safe first	5/23	1.586	1	3.32	7
Maximum immediate release	22/23	0.008696	0	0	0.2
Minimum unique-carrier exposure	17/23	0.2303	0	0.2686	3
Declared source order	11/23	0.6965	0.2857	2	3
Random-order deletion	8/23	1.384	0.6667	3.3	7
32-start random deletion	23/23	0	0	0	0
<i>Family B: HiGHS-OPTIMAL, exactly rechecked reference</i>					
Weighted greedy	82/111	0.02982	0	0.06452	0.5
Cardinality greedy	24/111	0.8904	0.25	2	7.5
Weighted greedy + reverse deletion	103/111	0.01038	0	0	0.5
Heaviest safe first	94/111	0.09983	0	0.2	3
Lightest safe first	11/111	4.128	0.7143	15	63
Maximum immediate release	94/111	0.09983	0	0.2	3
Minimum unique-carrier exposure	98/111	0.04435	0	0.05556	1
Declared source order	36/111	1.533	0.1818	5	20
Random-order deletion	28/111	1.753	0.2222	6.8	17
32-start random deletion	76/111	0.1324	0	0.5	2.667

Table 2 Registered heuristic burden quality. Relative gap is $(W - W^*)/W^*$. Attainment denominators include every registered row with an available reference; gap moments use exactly feasible returned libraries. Family A and Family B have different reference evidence and are deliberately not pooled.

Size	Preprocess n	Median var. red.	Median row red.	Median log PAR-2 ratio	32-start complete	Improved
N64-R32	28/32	1	1	1.187	28/32	20/28
N64-R64	28/32	1	1	1.29	28/32	19/28
N192-R32	28/32	1	1	2.031	15/32	12/15
N192-R64	28/32	1	1	2.243	27/32	21/27

Table 3 Registered preprocessing and multi-start evidence on structured instances. Reductions are shares removed. The log ratio is $\log(\text{PAR2}_{\text{full}}/\text{PAR2}_{\text{mandatory}})$, so a positive value favors the mandatory-only pipeline in elapsed penalized runtime despite the model-size reduction. Unavailable multi-start endpoints are not coded as zero improvement.

7.4 Preprocessing and realized design limits

Full fixed-point preprocessing has a strong realized effect. In every structured size cell, median variable and requirement reduction is complete among available records. This is evidence that forced-carrier propagation is important for these generators. It is simultaneously a limitation: most structured instances do not exercise a difficult residual MIP or DP search, so the study cannot identify general residual-search scaling.

The registered generator also contains infeasible requested cells. All 352 run units arising from 16 structured source-instance cells are retained, and the artifact audit shows that the failures remove complete combinations of low density and bundle/unique-

carrier settings rather than occurring at random. Effects and interactions requiring those absent cells are not estimable from this design.

Multi-start deletion completes all registered starts on 143 instances and improves on the first start on 96. Memory-limited primary runs remain failures; successful sensitivity records are not substituted into their denominators.

7.5 Limits of the comparison

Recorded times include first-call Julia compilation and contention among eight one-thread workers on the recorded worktree. They do not support steady-state speed rankings. The registered performance profile also mixes feasible-return and completed-search targets, so it is not comparative evidence here. Registered model outputs do not support structural inference: their HC3 uncertainty ignores repeated observations within instances, and entire requested generator cells are unavailable. Neither model refitting nor missing outcomes are used to improve the reported comparison.

Every unsuccessful run remains in the full denominator, including generation failures, memory limits, interruptions, absent primals, and nonbinary returned vectors. No run ended at the time limit. Online Resource 1, Section S7 gives the design, complete aggregate failure census, and exact-agreement audit.

8 Financial evidence: point-in-time portfolio-library retention

The financial study asks whether innovation-safe compression can reduce the active validation burden of portfolio-strategy libraries while preserving a larger menu of admissible future research options than a budget-matched frontier-only library. It is a retrospective development-facing panel, not an independent confirmatory market test. Its contribution is the combination of point-in-time source construction, an explicit portfolio grammar, protected proposal decisions, exact structural rechecks, and a transparent held-out null result.

8.1 Sequential information boundary

At each annual origin y , early historical blocks construct the source portfolio-strategy library and its operating frontier; the subsequent predecision block estimates compression profiles and solves the safe and frontier-only arms; year $y + 1$ enumerates closure-enabled proposal options and freezes both policy choices; and year $y + 2$ supplies the held-out portfolio return paths. Proposal outcomes do not enter compression. Evaluation outcomes do not enter compression, option generation, uncertainty estimation, adoption, or any policy choice. The full candidate ledger retains unavailable and rejected candidates rather than conditioning the analysis on successful selection.

The primary universe is formed from the most liquid eligible US common equities at each compression origin. A separate plain-ETF universe is the registered replication when its point-in-time eligibility gate passes. Eligibility never requires survival into the proposal or evaluation window; observed delisting returns are retained and a terminated position subsequently moves to cash. The single failed ETF universe gate

Universe	Origins	Mean source burden	Mean safe burden	Mean reduction	Added options	Held-out CE difference
Common equity	19	8169.2	2183.8	73.1%	65.95	0.0000
Plain ETF	18	8909.2	2231.7	74.7%	66.06	-0.0022

Table 4 Point-in-time portfolio-library retention results. Burden is the registered validation-computation index, not currency. Added options are safe-library proposal actions absent from the budget-matched frontier-only library. The held-out column is innovation-safe minus frontier-only certainty-equivalent difference: all 19 common-equity choices coincide, while one of 18 complete ETF choices differs. Safe solutions pass exact original-instance frontier, closure, inactive-retention, and burden rechecks; solver-reported optimality remains solver evidence.

remains in the registered denominator. Raw security identifiers, prices, and returns remain inside the licensed boundary.

Each source library applies a fixed factorial grammar of portfolio-level signals, filters, holding horizons, allocation rules, gross-exposure targets, and exits to the same contemporaneous universe. Closure consists of 31 declared rule, interface, data, corporate-action, portfolio-aggregation, and cost/capacity capabilities. These declarations are more operationally concrete than free-form module tags, but they remain a modeled grammar: they do not prove that tacit knowledge or every institutional artifact is frictionlessly recoverable.

8.2 Compression and protected proposal policy

The innovation-safe arm minimizes the registered validation-computation burden subject to exact source-frontier and full declared-closure equality. Its frontier-only comparator is augmented under the safe arm’s realized burden cap using a frozen preproposal order. The comparator action set is a subset of the safe action set in every gate-passed cell. Cash is always feasible, and the safe policy begins with the exact comparator choice. It can replace that choice only when a simultaneous moving-block lower confidence bound for incremental proposal value exceeds 0.0025 in annual certainty-equivalent return. The policy therefore treats closure as an option, not a forced trade.

Across the gate-passed origins, 3 safe solutions are resolved by exact preprocessing and 34 are returned with HiGHS `OPTIMAL` status. Every result is independently reconstructed and rechecked for mandatory inactive retention, exact table frontier equality, exact identity-closure equality, and exact integer burden. Those rechecks certify feasibility and burden; for the solver-resolved cells, global optimality remains mixed-integer solver evidence rather than a formal or exhaustive proof.

Table 4 gives the aggregate structural and held-out results. Innovation-safe retention reduces the mean source burden index by 73.1% in common equities and 74.7% in ETFs while preserving all registered capabilities. Relative to the budget-matched frontier-only library, it exposes on average 65.95 additional common-equity proposal actions and 66.06 additional ETF actions.

8.3 Held-out result: options are not exercises

The protected policy makes 0 common-equity replacements. Consequently, every common-equity held-out innovation-safe-minus-frontier-only contrast is exactly zero because the two policies choose the same action. This is not evidence that the two libraries are economically equivalent; it is evidence that none of the additional options clears the proposal-time uncertainty rule.

The ETF replication makes 1 replacement. That replacement realizes a held-out certainty-equivalent difference of -0.0390, producing a mean of -0.0022 over the complete ETF origins. All other complete ETF pairs choose the same action. The negative observation is retained and is not offset by reporting a favorable post hoc selector. Full origin-level results, the prespecified robustness grid, negative-control policies, search-bias diagnostics, and capacity records appear in Online Resource 1.

The contrast is a postdecision portfolio certainty-equivalent diagnostic under the registered risk and cost rule. It is not an alpha estimate, a forecast, or a deployable-performance claim. In particular, the many structural options do not count as independent financial observations, and the safe policy’s abstention cannot be reinterpreted as positive performance.

8.4 Robustness, licensed-data boundary, and managerial implication

The proposal robustness analysis retains the full prespecified grid, including unavailable cells. Common-equity safe and comparator choices remain identical throughout that grid. ETF differences remain confined to the same origin and disappear at higher transaction-cost settings. A closure-sham analysis that was not frozen before proposal outcomes became known is explicitly unavailable, not reconstructed post hoc. Capacity analysis scores frozen choices without reselection and retains missing-source, incomplete-path, and hard participation- cap failures in its denominators.

Licensed daily panels were used locally to construct the point-in-time universes and portfolio paths. The public artifact redistributes only aggregate structural manifests, cryptographic seals, permitted summaries, and synthetic records. It contains no raw CRSP/WRDS rows, security identifiers, or return paths. Independent replication of the licensed stages therefore requires separate data access, whereas all published table values can be regenerated from the committed public-safe manifests.

For research governance, the result is useful precisely because it is not a success story about returns. A manager can remove substantial declared validation burden while exactly preserving current opportunities and a much larger research-option menu. Whether any added option should be exercised is a separate statistical decision, and a protected selector can rationally abstain. Removal from the active maintenance set also does not authorize deletion from legal, compliance, or reproducibility archives. A stronger empirical claim would require a newly locked market vintage or independent panel, documented real capability inventories, and monetary maintenance costs calibrated without using postdecision outcomes.

Regime	True margin	Oracle mean [95% CI]	Learned mean [95% CI]	Adoption
Adverse margin	-0.0270	0.0000 [0.0000, 0.0000]	-0.0000 [-0.0000, 0.0000]	0.05%
Null margin	0.0000	0.0000 [0.0000, 0.0000]	-0.0002 [-0.0003, 0.0000]	5.76%
Low positive dose	0.0025	0.0026 [0.0019, 0.0032]	0.0001 [-0.0001, 0.0002]	7.35%
Powered positive dose	0.0540	0.0541 [0.0529, 0.0554]	0.0512 [0.0500, 0.0525]	94.17%

Table 5 Calibrated closure-option mechanism experiment, with 4096 synthetic worlds per prespecified regime. Current frontier and burden are equal by construction; only the innovation-safe library can admit the designated bridge descendant. Intervals are registered finite-simulation uncertainty summaries. This is synthetic mechanism evidence, not a market-return, alpha, forecasting, or causal estimate.

8.5 Calibrated closure-option mechanism experiment

The theorem is universal under its assumptions, but it does not say whether a finite-data policy can recognize a valuable bridge. We therefore isolate that question in a design-locked synthetic experiment. The innovation-safe and frontier-only libraries have identical current frontiers and identical retention burdens. Only the safe library carries the capability needed to run one declared research project. For delay (d), survival probability ρ , admission probability π , descendant gain G , and research cost κ , the registered treatment margin is

$$M = \beta^d \rho^d \pi G - \kappa.$$

The safe oracle researches exactly when $M > 0$; the frontier-only project is infeasible. A learner observes an independent noisy proposal signal and adopts only when it exceeds a threshold calibrated under the null.

The noise law uses only aggregate nuisance estimates from 2479 complete proposal paths and 623647 within-cell residual observations in the point-in-time financial study. Cross-strategy date means are removed before estimating volatility, lag-one dependence, and tail thickness. These quantities calibrate noise and dependence, not the sign of any regime. The test-world seeds are disjoint from calibration, and the powered margin is fixed at twice the null threshold before test simulation. Thus its size is a power-design dose, not an estimated return premium.

Table 5 reports all four prespecified regimes. At the low positive dose, the oracle value is positive but the learned-value interval crosses zero and adoption is only 7.35%. At the powered dose $M = 0.0540$, the learned mean is 0.0512 and adoption is 94.17%. The null point adoption rate, 5.76%, satisfies the registered 6.5% finite-test gate; its Wilson upper endpoint is 6.52%, so the interval is not wholly below that threshold. Every world passes the equal-frontier, equal-burden, and bridge-availability structural checks.

This evidence makes the option/exercise distinction concrete. Exact closure preservation can create access to a genuinely positive project while a conservative learner rationally fails to exercise a small option at the available signal-to-noise ratio. Conversely, the powered row shows recovery only for a deliberately larger treatment. The experiment validates the declared bridge mechanism under calibrated synthetic noise.

It is not a historical market test, a causal return estimate, an alpha estimate, or a claim that the powered margin is empirically prevalent.

9 Relation to existing work

The financial motivation connects costly information acquisition with the maintenance of productive capabilities. [Van Nieuwerburgh and Veldkamp \(2010\)](#) jointly model information acquisition and portfolio choice, showing how learning incentives affect investment. [Eisfeldt and Papanikolaou \(2013\)](#) model organization capital embodied in key talent and link its firm-specific productivity and outside options to financial risk. Our retention problem focuses on declared reusable inputs to later strategy research: removing a carrier can change which projects remain feasible even when its current operating contribution is redundant. The closure is a specified capability inventory, not an estimate of tacit knowledge or an institutional measure of organization capital. This paper asks when a smaller library preserves that inventory and its modeled uses; it does not determine whether every capability is worth maintaining.

Belief-state reduction supplies the control architecture used here. [Smallwood and Sondik \(1973\)](#) and [Sondik \(1978\)](#) develop finite- and infinite-horizon partially observed control; [White \(1976\)](#) treats a partially observed semi-Markov process. Our additional question concerns the library component of state: which retained strategies leave both operating choices and the declared generation process unchanged? The frontier–closure pair summarizes that component under the factorization assumption. The belief and active-project variables remain in the controlled state.

The preservation target distinguishes this problem from nearby reductions. Incremental pruning removes redundant value vectors in a fixed POMDP backup ([Cassandra et al. 1997](#)); state aggregation identifies behaviorally equivalent states for a specified model ([Givan et al. 2003](#)). Neither requires a removed strategy to remain available as a carrier for subsequent generation. Compact control libraries trade static usefulness against library size and search effort ([Dey et al. 2012](#)). Reward-free policy compression uses set cover to represent state–action distributions for downstream tasks ([Mutti et al. 2022](#)). Here the choices are subsets of a given source, and the preserved objects are its operating frontier and generative closure.

Once identity closure makes those objects a finite tagged requirement set, the algorithmic core is classical weighted cover. The complexity reduction uses Karp’s set-cover problem ([Karp 1972](#)), and the greedy guarantee is transferred from [Chvátal \(1979\)](#). The modeling contribution is the mapping from the financial preservation condition to that problem; the generic cover algorithms are established methods. Nonidentity closure requires a separate representation of capability interactions. Maximum-coverage or adaptive selection guarantees do not follow merely from calling capabilities a library.

In finance, automated construction has long used genetic search and reinforcement methods ([Allen and Karjalainen 1999](#); [Moody and Saffell 2001](#)). Those studies concern constructing or operating strategies; our decision is which entries in a maintained source can be removed while preserving its specified opportunities. Our burden index measures the declared maintenance requirements; it is not an estimate of institutional compliance costs.

A larger candidate menu also creates a larger selection problem. The literature on data snooping and multiple testing explains why more available strategies cannot itself establish better investment performance (Sullivan et al. 1999; White 2000; Harvey et al. 2016). The financial study therefore freezes a simultaneous proposal-time adoption rule before held-out evaluation and reports abstention and adverse exercise. Its known historical setting still limits external validity, and post-publication decay provides a further reason for restraint (McLean and Pontiff 2016).

10 Conclusion

A strategy can be redundant for current operation while remaining useful as a carrier of a future research capability. A manager who wants to reduce active maintenance burden must specify both roles before deleting entries. Under the declared finite factorization, preserving the operating frontier and generative closure preserves the library’s productive process. The theorem’s assumptions are substantive: dependence on a named strategy or another omitted attribute can invalidate that reduction.

Under identity closure, minimum-burden safe retention is a tagged weighted cover. This gives access to established exact and approximate methods, with independent checks in the original strategy coordinates. Safe local deletion alone provides no general burden guarantee, as the heaviest-first family shows. The registered benchmark confirms exact-method agreement where available and documents burden quality and preprocessing effects within its realized design; it does not support general speed rankings.

The point-in-time application preserves many research options at lower declared burden, yet the protected common-equity policy makes no additional investment and its one ETF replacement loses held-out value. The synthetic experiment shows that small positive bridge value can also be difficult to detect under the specified noise. Preserving an option therefore leaves a separate statistical and economic decision about exercise.

The natural extensions are capability interactions beyond identity closure and an independent financial study with documented institutional capabilities and maintenance costs measured before future outcomes. The present findings support a precise retention contract, rather than a claim of alpha or deployable performance.

Supplementary information

Online Resource 1, *Supporting Proofs, Methods, and Reproduction*, contains supporting proofs and counterexamples, complete benchmark outcome accounting, financial and synthetic methods, origin-level results, and a reproduction guide. The combined preprint appends it with S-prefixed numbering. The editable package includes the document inputs and the public evidence used here.

Statements and Declarations

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Data availability. The accompanying source package includes exact tables, public aggregate financial outputs, benchmark records, and the synthetic experiment ledger. Raw and row-level CRSP/WRDS observations are excluded; reproducing the licensed stages requires independent licensed access. The reproduction guide identifies the required inputs and source files.

Code availability. Julia 1.12.6 and Lean 4.32.0 sources and pinned environments are available at <https://github.com/dramirez37/financial-strategy-library-compression>. The [versioned preprint package](#) contains the PDFs, editable source, and public evidence. Its reproduction manifest binds the selected scientific sources by hash. The document archive builds the PDFs independently; full computational replication uses the specified repository sources. Original software is MIT licensed and manuscript material is CC BY 4.0, subject to the licensed-data and third-party exclusions in `LICENSE`.

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Online Resource 1: Supporting Proofs, Methods, and Reproduction

for “Innovation-Safe Compression of Financial Strategy Libraries: Semantics, Complexity, and Algorithms”

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Abstract

This resource supplies the supporting probability and preservation arguments, covering and complexity proofs, algorithm guarantees, and deletion counterexample for the accompanying article. It also gives complete benchmark outcome accounting, financial and synthetic methods, origin-level financial results, verification scope, and instructions for reproducing the paper and its evidence. Public data are distributed separately in machine-readable form; licensed CRSP/WRDS observations require independent access.

Version: v0.3.0-ssrn.

Evidence boundary: Human proof, Lean kernel verification, exact finite computation, solver evidence, synthetic evidence, and retrospective financial evidence are identified separately.

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S1 Exact probability normalization and projection identities

For a finite carrier A , every probability law in the model is an exact rational table

$$\Delta_{\mathbb{Q}}(A) := \left\{ p : A \rightarrow \mathbb{Q} : p(a) \geq 0 \text{ for every } a, \sum_{a \in A} p(a) = 1 \right\}.$$

Beliefs, belief-kernel rows, and candidate-generation rows use this convention. At fixed project q , belief b , and closure C , verification assigns

$$\Gamma(q, b, C)(s) = G(q, b, C)(s) \nu(q, b, C, s), \quad s \in S, \quad (\text{S1.1})$$

$$\Gamma(q, b, C)(\text{none}) = G(q, b, C)(\text{none}) + \sum_{s \in S} G(q, b, C)(s) (1 - \nu(q, b, C, s)). \quad (\text{S1.2})$$

Every term is nonnegative. Summing (S1.1)–(S1.2) cancels the verification split and recovers the unit mass of G , so Γ is normalized.

Conditional on initial belief b , productive state $K = (F, C)$, and project q of duration d_q , let $\Lambda_q(\mathbf{b}, o \mid b, K)$ be the exact joint law of the belief path $\mathbf{b} = (b_0, \dots, b_{d_q})$ and admitted outcome o . Its specified marginals are

$$\sum_o \Lambda_q(\mathbf{b}, o \mid b, K) = \mathbf{1}\{b_0 = b\} \prod_{t=0}^{d_q-1} P_B(b_t, b_{t+1}), \quad (\text{S1.3})$$

$$\sum_{\mathbf{b}} \Lambda_q(\mathbf{b}, o \mid b, K) = \Gamma(q, b, C)(o). \quad (\text{S1.4})$$

These equations do not impose independence between the belief path and the admitted outcome.

For a raw source library L , deterministic update gives

$$\mathcal{Q}_q^{\text{raw}}(b', L' \mid b, L) := \sum_{\substack{\mathbf{b}, o: b_{d_q} = b' \\ L \oplus o = L'}} \Lambda_q(\mathbf{b}, o \mid b, K_L).$$

For realizable $K = (F, C)$, the two compressed pushforwards are

$$\overline{T}_q(K' \mid b, K) := \sum_{o: \text{addK}(K, o) = K'} \Gamma(q, b, C)(o), \quad (\text{S1.5})$$

$$\overline{\mathcal{Q}}_q(b', K' \mid b, K) := \sum_{\substack{\mathbf{b}, o: b_{d_q} = b' \\ \text{addK}(K, o) = K'}} \Lambda_q(\mathbf{b}, o \mid b, K). \quad (\text{S1.6})$$

Finite deterministic pushforward preserves nonnegativity and total mass. The local update identity $K_{L \oplus o} = \text{addK}(K_L, o)$ also makes every compressed successor realizable. These identities supply the normalization clause of the raw-to-compressed projection theorem; they do not supply the factorization, positive-duration induction, or contraction arguments by themselves.

S1.1 Complete projection proof

The article assumes a finite hidden state, finite filtered belief carrier, finite strategy catalog, finite project catalog, positive integer project durations, exact rational transition and completion laws, and a mandatory inactive strategy. Every operating reward, project availability indicator, project cost, duration, generation law, verification law, and joint terminal belief-admission law factors through the current belief and the productive library state $K_L = (F_L, C_L)$. Raw insertion and compressed insertion obey

$$K_{L \oplus o} = \text{addK}(K_L, o). \quad (\text{S1.7})$$

Stationary claims additionally assume a common discount factor $0 < \beta < 1$; finite-horizon claims do not.

Theorem S1.1 (Raw-to-compressed controlled semi-Markov projection). *Under the preceding assumptions, $K_L = (F_L, C_L)$, augmented during an unfinished project by project identity and remaining duration, is a controlled semi-Markov state for the raw library process. Raw and compressed finite-horizon values agree at every raw state, and the complete sets of maximizing actions agree under the natural action identification. Under the stationary discount clause, the raw and compressed Bellman operators have unique fixed points with the same values and maximizing stationary-action correspondence.*

Proof Fix a raw library L , belief b , and feasible action. For Continue, the operating reward and belief transition depend on L only through F_L , hence through K_L . For a project q , the factorization assumption makes feasibility, cost, duration, the incumbent operating-reward block, and the joint completion law functions of (b, K_L, q) . Equations (S1.1)–(S1.4) show that this law is normalized without imposing independence between the belief path and admission. Pushing it through the deterministic raw update and using (S1.7) gives exactly the compressed transition law (S1.6). Its support consists of realizable compressed states. Thus two raw libraries in the same K -fiber have the same feasible actions, current reward blocks, durations, and conditional laws of the next compressed decision state.

For a finite calendar horizon h , prove equality of every corresponding action value by strong induction on h . The claim is immediate at horizon zero. Continue invokes the induction hypothesis at $h - 1$; a feasible project invokes it after its positive duration d_q , at $h - d_q < h$. Current rewards and transition laws already agree by the preceding paragraph. Each corresponding action value therefore agrees, as do the finite maxima and the complete maximizing-action sets. Finiteness makes every maximum attained.

For the stationary clause, cast the exact rational tables into the reals and use the sup norm on the finite state carrier. Conditional expectation is nonexpansive. Continue discounts continuation by β , and a project discounts it by $\beta^{d_q} \leq \beta$. Maximization over the common

finite action set preserves this modulus, so both Bellman operators are β -contractions. Bellman intertwining, established by the action-value identity above, maps the compressed fixed point to a raw fixed point. Banach uniqueness gives value equality; the same pointwise action-value identity gives the complete maximizing stationary-action correspondence. No closure detectability assumption enters this forward projection. $\square$

S2 Productive Preservation and the Frontier–Closure Quotient

For a finite library A containing the inactive strategy, write

$$F_A(b) = \max_{s \in A} j_s(b), \quad C_A = \text{cl} \left(\bigcup_{s \in A} \text{mods}(s) \right), \quad K_A = (F_A, C_A).$$

The forward preservation result uses only the process factorization and local update assumptions stated in Section S1. A converse needs a distinct detectability assumption: whenever two libraries have equal frontiers but unequal closures, at least one availability-tagged project cost, duration, incumbent-reward block, or joint terminal observation differs after the declared generation, verification, admission, and projection steps.

Theorem S2.1 (Frontier–closure characterization). *If $F_A = F_L$ and $C_A = C_L$, then A and L induce identical productive observations and identical finite-horizon values and maximizing actions. The stationary conclusions of Section S1 also agree under its discount clause. If closure is detectable in the preceding sense, equality of all productive observations together with frontier equality implies $C_A = C_L$.*

Proof Frontier–closure equality is $K_A = K_L$. The factorization assumption then identifies project availability, cost, duration, incumbent operating rewards, generation and verification laws, and the conditional joint completion law. The normalized admission map and deterministic pushforward from Section S1 therefore identify the complete productive transition observations. The finite-horizon and stationary value and action claims follow from Theorem S1.1. Conversely, suppose the frontiers and all named productive observations agree. If the closures differed, detectability would produce a named observation that differs, a contradiction. Hence the closures agree. Detectability is used only for this reverse implication. $\square$

Corollary S2.2 (Source-relative safe compression). *For $s_0 \in A \subseteq L$, exact equality $F_A = F_L$ and $C_A = C_L$ is sufficient for productive preservation. With exact positive active weights and $w_{s_0} = 0$, the finite safe-feasible family contains L and therefore has at least one minimum-burden member.*

Proof The productive claim is the forward direction of the theorem. The source is safe relative to itself. A finite source has finitely many sublibraries, so the nonempty feasible set has an attained minimum under exact weights. $\square$

The converse characterization is deliberately restricted. It is not a claim that K is a generic coarsest bisimulation or a full-abstraction theorem. It states that the declared finite process factors through K , and that the additional detectability clause is sufficient to recover closure equality from the named productive observations.

The operating observation is the maximum modeled payoff of one catalog entry. An entry may already specify a portfolio, but the result does not preserve an unmodeled allocation across several entries. Such an application requires a preservation target containing the relevant joint payoff information and feasible allocations before the factorization argument can be applied.

S2.1 Operational factorization audit

The assumption can be checked against a proposed application by asking whether, conditional on belief and active-project state, every operating reward, generation law, verification outcome, admission probability, duration law, completion observation, and next-library update depends on the raw library only through (F_A, C_A) . Provenance, cardinality, identity-specific switching cost, or a requirement for a named parent violates the assumption unless it is added to the compressed state. A finite module-gated construction satisfies the assumption when all projects are indexed only by requirements contained in C_A , incumbent rewards use only F_A , and all subsequent kernels use only belief, active-project variables, and those two objects.

The following exact countermodel shows why the checklist is substantive. Let s_0 be inactive and let active strategies a and b have identical zero profiles and the same single module. Then libraries $\{s_0, a\}$ and $\{s_0, b\}$ have identical frontier and closure. If a project is available only when the raw identity a is present, their project menus differ. The model therefore does not factor through (F, C) , and the projection theorem makes no claim about it. The countermodel changes no theorem: it falsifies an omitted assumption, not the stated conditional result.

S3 Tagged Cover, Complexity, and Boundary Counterexamples

This section gives the complete proofs supporting the article’s identity-closure covering representation and complexity result. The semantic safe-feasibility condition is defined for arbitrary declared closure; only the tagged row reduction and the algorithms built on it assume identity closure.

S3.1 An exact tagged covering representation under identity closure

Fix a finite belief set B , a finite module carrier M , a finite strategy catalog S , and a source library $L \subseteq S$. Every admissible library contains the distinguished inactive strategy s_0 . Profiles are exact rational functions $j_s : B \rightarrow \mathbb{Q}$, $j_{s_0}(b) = 0$ for every b ,

and $\text{mods}(s_0) = \emptyset$. Each $\text{mods}(s)$ is a subset of M . For an admissible library A , let

$$F_A(b) := \max_{s \in A} j_s(b), \quad U_A^{\text{raw}} := \bigcup_{s \in A} \text{mods}(s), \quad C_A := \text{cl}(U_A^{\text{raw}}).$$

The maximum is attained because A is finite and nonempty. The inactive strategy also implies $F_A(b) \geq 0$.

In this subsection only, closure is the identity on 2^M . Thus $C_A = U_A^{\text{raw}}$ for every A . Define the source-relative tagged requirement universe

$$\mathcal{U}_L^{\text{tag}} := \{\text{frontier}(b) : b \in B\} \dot{\cup} \{\text{module}(m) : m \in C_L\}. \quad (\text{S3.1})$$

The constructors are disjoint: a frontier tag and a module tag are different even if their underlying identifiers have the same printed label. For each source strategy $s \in L$, define

$$\mathcal{R}_s := \{\text{frontier}(b) : j_s(b) = F_L(b)\} \cup \{\text{module}(m) : m \in \text{mods}(s)\}. \quad (\text{S3.2})$$

Equality in the first set is exact. Consequently every tied source maximizer covers the frontier row; no tie-breaking rule or distinguished maximizer enters the construction.

Theorem S3.1 (Tagged-cover equivalence under identity closure). *Under the assumptions above, let L' be an admissible source sublibrary: $s_0 \in L' \subseteq L$. Then*

$$F_{L'} = F_L \text{ and } C_{L'} = C_L \iff \mathcal{U}_L^{\text{tag}} \subseteq \bigcup_{s \in L'} \mathcal{R}_s. \quad (\text{S3.3})$$

All belief rows are retained in (S3.3). If $F_L(b) = 0$, the mandatory inactive strategy attains that frontier and automatically covers $\text{frontier}(b)$.

Proof Suppose first that $F_{L'} = F_L$ and $C_{L'} = C_L$. Fix $b \in B$. The finite maximum defining $F_{L'}(b)$ has an attainer $s \in L'$. Frontier equality gives $j_s(b) = F_{L'}(b) = F_L(b)$, so $\text{frontier}(b) \in \mathcal{R}_s$. This argument is existential and therefore permits any number of tied attainers. Next fix $m \in C_L$. Identity closure and closure equality give

$$m \in C_{L'} = U_{L'}^{\text{raw}} = \bigcup_{s \in L'} \text{mods}(s).$$

Hence some $s \in L'$ carries m , so $\text{module}(m) \in \mathcal{R}_s$. Every tagged requirement is covered.

Conversely, suppose every tagged requirement is covered. Because $L' \subseteq L$, maximization over the smaller set gives $F_{L'}(b) \leq F_L(b)$ for every b . Coverage of $\text{frontier}(b)$ supplies some $s \in L'$ with $j_s(b) = F_L(b)$. Therefore $F_L(b) = j_s(b) \leq F_{L'}(b)$, proving frontier equality pointwise. This also covers the zero-frontier case: $s_0 \in L'$ is itself a valid attainer.

For modules, source inclusion implies $U_{L'}^{\text{raw}} \subseteq U_L^{\text{raw}}$. If $m \in U_L^{\text{raw}} = C_L$, coverage of $\text{module}(m)$ supplies a retained carrier, so $m \in U_{L'}^{\text{raw}}$. Thus the two raw unions are equal. Identity closure converts this equality into $C_{L'} = C_L$. A module may have one or many source carriers; the row constraint requires at least one and does not select a canonical carrier. $\square$

Weighted binary formulation.

Let the exact rational retention weights satisfy $w_{s_0} = 0$ and $w_s > 0$ for $s \neq s_0$. Introduce one variable $x_s \in \{0, 1\}$ for each $s \in L$, where $x_s = 1$ means that s is retained. The theorem gives the exact formulation

$$\min_x \sum_{s \in L} w_s x_s, \quad (\text{S3.4})$$

$$\text{s.t.} \quad \sum_{s \in L: j_s(b) = F_L(b)} x_s \geq 1, \quad b \in B, \quad (\text{S3.5})$$

$$\sum_{s \in L: m \in \text{mods}(s)} x_s \geq 1, \quad m \in C_L, \quad (\text{S3.6})$$

$$x_{s_0} = 1, \quad x_s \in \{0, 1\}, \quad s \in L. \quad (\text{S3.7})$$

The source-sublibrary restriction is built into the column set L . The selected library is $L_x = \{s \in L : x_s = 1\}$, its burden is exactly the objective in (S3.4), and Theorem S3.1 identifies its feasible vectors with the exact innovation-safe sublibraries. Zero-frontier rows are redundant after $x_{s_0} = 1$, but retaining them makes the representation a literal tagged copy of every declared belief and avoids a hidden positive-frontier convention.

Why arbitrary closure is different.

For a nonidentity closure operator, equality of closed unions need not be equivalent to retaining a raw carrier for each member of the source closure. Capabilities may be generated jointly. A correct general-closure model must enforce the set-valued equality

$$\text{cl} \left(\bigcup_{s: x_s = 1} \text{mods}(s) \right) = C_L$$

or use a separately proved valid extended formulation. The independent module rows in (S3.6) have no automatic validity in that setting.

Exact boundary examples.

The following finite examples use integer profiles and exact finite sets.

1. *Frontier rows alone are insufficient.* Take one belief, one module m , identity closure, and $L = \{s_0, s_m\}$, with both profiles zero, $\text{mods}(s_0) = \emptyset$, and $\text{mods}(s_m) = \{m\}$. The sublibrary $\{s_0\}$ covers the only frontier row and preserves the zero frontier, but its closure is empty rather than $\{m\}$.
2. *Module rows alone are insufficient.* Take one belief, identity closure, and $L = \{s_0, s_1\}$, with $j_{s_0} = 0$, $j_{s_1} = 1$, and both module sets empty. The source module row family is empty, so $\{s_0\}$ satisfies it vacuously, but its frontier is zero rather than one.

3. *Independent raw-module rows can fail under closure complementarity.* Let $M = \{a, b, c\}$ and

$$\text{cl}(A) = \begin{cases} A \cup \{c\}, & \{a, b\} \subseteq A, \\ A, & \text{otherwise.} \end{cases}$$

This map is extensive, monotone, and idempotent. Let the zero-profile source be $L = \{s_0, s_a, s_b\}$, where s_a carries only a and s_b carries only b . Then $C_L = \{a, b, c\}$, and the source is of course safe-feasible relative to itself. Nevertheless a raw-carrier row for $c \in C_L$ has no covering source strategy. The naive independent-row model therefore rejects even the source library.

The deterministic machine-readable realization is `journal/aor/fixtures/tagged_cover_equivalence_v1.json`. It records exact finite computation in Julia, including all sublibraries of its declared small-library grid; it is neither an exhaustive proof over arbitrary finite instances nor a solver optimality certificate. The universal core biconditional and exact burden identity are separately checked in the Lean source `TaggedSafeCompression.lean` under `formal/StrategyInnovation/Optimization/`. Those declarations formalize neither a complexity result nor an approximation guarantee.

S3.2 Computational complexity of exact compression

We use an explicit finite encoding. A source instance lists a finite strategy library $L = \{s_0, s_1, \dots, s_n\}$, a finite nonempty belief set $\mathcal{B}$, exact binary-rational profiles $j_s(b)$, a finite module universe $\mathcal{M}$, and the incidence rows $\text{mods}(s) \subseteq \mathcal{M}$. Strategy s_0 is the mandatory inactive strategy, has profile zero at every belief, carries no modules, and has zero burden. Every active strategy has a positive binary-integer weight w_i , and the threshold B is a nonnegative binary integer. Closure is identity closure,

$$C(A) = \bigcup_{s \in A} \text{mods}(s).$$

The source frontier is $F_L(b) = \max_{s \in L} j_s(b)$, and the burden of a sublibrary is $W(A) = \sum_{s_i \in A, i \geq 1} w_i$. The decision problem **IDENTITY-CLOSURE SAFE COMPRESSION (ISC-ID)** asks whether there is an $A \subseteq L$ such that $s_0 \in A$, $W(A) \leq B$, $F_A = F_L$ at every listed belief, and $C(A) = C(L)$.

The hardness source below is a deliberately normalized version of weighted set cover. In **FULL-UNION WEIGHTED SET COVER (FU-WSC)**, the input is a nonempty finite universe U , an indexed finite family $(A_i)_{i=1}^n$ whose union is U , positive binary-integer costs c_i , and a nonnegative binary-integer budget B . The question is whether some index set $I \subseteq \{1, \dots, n\}$ satisfies $\bigcup_{i \in I} A_i = U$ and $\sum_{i \in I} c_i \leq B$. Encodings that fail these stated syntactic restrictions are rejected, so FU-WSC is a language rather than an unverified promise problem.

Lemma S3.2 (Normalized weighted set cover). *FU-WSC is NP-complete.*

Proof Input validity is checkable in polynomial time. A selected-index bit vector is a polynomial certificate: unioning its listed incidence rows and adding its binary-integer costs verifies feasibility in polynomial time. Thus FU-WSC is in NP.

For hardness, reduce the standard SET COVER decision problem, one of Karp's NP-complete covering problems (Karp 1972). Write an input as $(U, (A_i)_{i=1}^n, k)$, with nonnegative cardinality bound k . If $U = \emptyset$, output the fixed yes-instance consisting of universe $\{u\}$, the one set $\{u\}$, cost one, and budget one. If $U \neq \emptyset$ but $\bigcup_i A_i \neq U$, output the fixed no-instance with universe $\{u\}$, the one set $\{u\}$, cost two, and budget one. Otherwise retain the universe and family, assign every set unit cost, and set the budget to k . Every output has a nonempty universe, a full-union family, positive integer costs, and a nonnegative integer budget. The first branch preserves the fact that the empty universe is coverable; the second preserves the fact that an uncovered element makes the source instance a no-instance; and in the third branch cost is exactly selected-set cardinality. The case split, union check, and copied incidence representation have polynomial size and construction time. Hence the mapping preserves yes- and no-instances and proves NP-hardness. $\square$

Theorem S3.3 (Identity-closure safe-compression complexity). *ISC-ID is NP-complete. More specifically, it remains NP-complete on the subclass with exactly one belief, zero profiles for every active strategy, an inactive strategy with zero profile, zero burden, and no modules, and a source library whose active module rows have full union. The result holds even when every active weight is one. Consequently, minimum-burden identity-closure safe compression is NP-hard. Any broader safe-compression model class containing this identity-closure subclass is NP-hard; no membership claim for an arbitrary nonidentity closure operator follows without a polynomial closure-equality verifier in its declared encoding.*

Proof Membership in NP. Encode a proposed retained library by one bit per source strategy. Inactive retention and source inclusion are direct bit checks. Binary-integer addition checks $W(A) \leq B$. For every explicitly listed belief, exact rational comparison finds the source and candidate maxima and checks their equality. Under identity closure, bitwise union of the listed module-incidence rows checks $C(A) = C(L)$. Integer and rational operations have bit complexity polynomial in the explicit input length. Thus ISC-ID is in NP.

Reduction algorithm. Given a valid FU-WSC instance $(U, (A_i)_{i=1}^n, (c_i)_{i=1}^n, B)$, create one module m_u for each $u \in U$, one active strategy s_i for each set A_i , and one mandatory inactive strategy s_0 . Use one belief b_0 . Set

$$j_{s_i}(b_0) = j_{s_0}(b_0) = 0, \quad \text{mods}(s_i) = \{m_u : u \in A_i\}, \quad \text{mods}(s_0) = \emptyset.$$

Give s_i weight c_i , give s_0 weight zero, take the complete constructed library as the source, use identity closure, and copy the budget without alteration.

Polynomial construction size. The construction adds one module per universe element, one active strategy per set, one inactive strategy, and one belief. Its module-incidence table is the set-cover incidence table; the remaining profiles are $n + 1$ zero entries, and costs and budget are copied. A dense encoding has size $O(n|U|)$ plus the copied identifier and integer bit lengths, while a sparse encoding is linear in the copied incidence list plus this overhead. Either encoding is polynomial in the corresponding explicit source encoding.

Forward feasibility. Suppose I is a set cover with $\sum_{i \in I} c_i \leq B$. Retain $A = \{s_0\} \cup \{s_i : i \in I\}$. Both source and retained frontiers equal zero at b_0 , with s_0 itself attaining that value. Identity closure gives

$$C(A) = \bigcup_{i \in I} \text{mods}(s_i) = \{m_u : u \in U\} = \bigcup_{i=1}^n \text{mods}(s_i) = C(L),$$

where the last equality uses the full-union condition. The inactive strategy is retained, and $W(A) = \sum_{i \in I} c_i \leq B$, so A is a yes-certificate for the constructed ISC-ID instance.

Reverse feasibility. Suppose A is a safe constructed sublibrary within threshold, and let $I = \{i : s_i \in A\}$. Exact closure preservation and the empty inactive module row imply

$$\bigcup_{i \in I} \{m_u : u \in A_i\} = C(A) = C(L) = \{m_u : u \in U\}.$$

The module labels are in one-to-one correspondence with U , so $\bigcup_{i \in I} A_i = U$. Thus I is a set cover. Frontier preservation adds no hidden selection requirement because the mandatory inactive strategy already attains the common zero frontier.

Threshold preservation. The inactive strategy has zero burden, active weights are copied exactly, and the threshold is copied exactly. Therefore $W(A) = \sum_{i \in I} c_i$ for every selected index set I , not merely for optimal selections. Hence $W(A) \leq B$ if and only if the corresponding set cover has cost at most B . The reduction is a polynomial many-one reduction. Together with membership in NP and Lemma S3.2, this proves NP-completeness of the stated restricted subclass and therefore of ISC-ID. The normalization proof uses unit costs on its nontrivial branch, so the restricted subclass remains NP-complete with unit active weights. Solving the associated minimum-burden problem would decide whether its optimum is at most B , which proves NP-hardness of the optimization problem. $\square$

The encoding, normalization, and both directions above establish the human proof. Exact reduction fixtures are supplementary finite checks.

S4 Exact Preprocessing Propositions

Fix an identity-closure tagged covering instance after all declared mandatory columns have been selected. Let Q be the remaining requirement rows, A the available nonfixed columns, $R_s^Q = R_s \cap Q$, and let each optional weight be a strictly positive exact rational. There are no identity-sensitive side constraints, groups, exact-cardinality constraints, or mutual exclusions: this is the pure covering model. Every transformation records original identifiers, forced selections, fixed burden, equal-weight substitutions, and merged-row memberships.

Proposition S4.1 (Mandatory carrier and forced propagation). *If $u \in Q$ has exactly one carrier $s \in A$, every feasible completion selects s . Fixing s , adding w_s to the objective offset, deleting the rows it satisfies, and recomputing carriers preserves feasibility, the exact optimum value, every optimum identity, and reconstruction.*

Proof Any feasible completion must cover u , and s is its only available carrier, so s occurs in every completion. Removing already satisfied rows does not change the constraints on the remaining columns. Splitting every feasible objective into the fixed term w_s and its residual term is a bijection between feasible residual selections and feasible original selections containing the already forced columns. Repeating the argument after every selection proves propagation. $\square$

Proposition S4.2 (Coverage dominance). *Suppose $s, t \in A$, $s \neq t$, $R_s^Q \subseteq R_t^Q$, and $w_t \leq w_s$. If s is not mandatory, deleting s preserves feasibility, the optimum value, and at least one optimum. If $w_t < w_s$, no optimum contains s , so all optimizer identities are preserved. If $w_t = w_s$, all optimizer identities need not be preserved; a reconstruction record must retain the admissible substitution $s \leftrightarrow t$ where its surrounding selection remains feasible.*

Proof Take a feasible selection containing s . If it also contains t , then deleting s preserves coverage and strictly lowers burden because $w_s > 0$. If it does not contain t , replace s by t . Superset coverage preserves feasibility and the burden weakly falls. Thus every feasible solution has a no-worse representative without s , which proves feasibility and optimum-value preservation and supplies at least one reduced optimum. Under strict inequality, either case strictly improves any selection containing s , so no original optimum uses it. Under equality, replacement can exchange two distinct optimizer identities without changing burden. $\square$

Proposition S4.3 (Duplicate columns and empty contribution). *For columns with identical residual coverage, retaining a stable minimum-weight representative and deleting the others preserves feasibility, the exact optimum value, and at least one optimum. Strictly heavier duplicates cannot occur in an optimum. Equal-weight duplicates require an explicit substitution class to reconstruct all optimizer identities. A nonmandatory positive-weight column with empty residual coverage may be deleted; it occurs in no optimum.*

Proof Identical-coverage replacement leaves every row status unchanged and weakly reduces burden, with strict reduction for a heavier duplicate. Stable choice only fixes which representative remains in the reduced model. Recording the equal-weight class recovers the alternative identities consistent with the rest of a lifted selection. An empty-contribution column changes no residual constraint, and deleting it strictly lowers any selection that contains it. $\square$

Proposition S4.4 (Duplicate requirement rows). *If two residual requirement rows have identical carrier sets, replacing them by one representative preserves feasibility, optimum value, at least one optimum, and all optimizer identities. Recording both original row identifiers preserves the feasibility certificate.*

Proof A selection intersects one carrier set if and only if it intersects the identical other carrier set. The two inequalities therefore define the same feasible selections and do not enter the objective. Merging them changes no optimizer; the row-membership record merely expands the final certificate. $\square$

The implementation applies the rules to a stable fixed point and restarts its ordered scan after every transformation. Each nontrivial step removes a row or column or fixes a previously available column, so finiteness proves termination. The complete machine audit distinguishes forced columns, dominated columns, duplicate columns, empty columns, satisfied rows, and duplicate rows. These are exact combinatorial propositions.

Julia randomized exhaustive tests and Lean finite-mask identities are separate evidence layers; neither turns a downstream solver status into a formal optimality proof.

S5 Algorithm Proofs, Certificates, and Pseudocode

The following proofs concern validated tagged identity-closure instances. Every returned library is reconstructed to original identifiers and then checked against the original source frontier, source closure, mandatory set, and exact burden. Approximation statements do not extend to arbitrary closure.

S5.1 Independent exact algorithms

The tagged representation gives two independent exact finite methods. They share only validated input, mandatory-selection preprocessing, and the final source-library certificate; their search procedures are different. Complete enumeration is the small-library oracle. Requirement-mask dynamic programming is parameterized by the number of residual tagged requirements.

Fix every mandatory strategy first. Let W_0 be their exact burden, delete every requirement they satisfy, and let the residual instance have strategies $1, \dots, n$ and requirements $1, \dots, r$. Write $R_i \subseteq [r]$ for the residual tagged coverage of strategy i and $w_i > 0$ for its exact weight. Mandatory preprocessing alone removes no optional strategy, so it preserves all optimizer identities. If a serialized instance already contains a valid stronger preprocessing map, the algorithms use its residual model, exact offset, and declared reconstruction map; complete-tie output is rejected when that map says that some equal-cost identities are unreconstructable.

The preprocessing assumptions and reconstruction qualifications are proved in Section S4.

S5.1.1 Complete enumeration

The enumeration method visits every one of the 2^n residual binary selections. It tests tagged coverage directly, sums burden in exact rational arithmetic, and retains either every minimum or the declared stable representative. It then lifts forced selections and rechecks mandatory retention, the original operating frontier, the original identity closure, and the exact burden independently of the search test.

Proposition S5.1 (Correctness of complete enumeration). *On a valid finite tagged identity-closure instance, complete enumeration returns a minimum-burden safe source sublibrary. When complete-tie retention is requested and the preprocessing map preserves all optimizer identities, it returns every minimum-burden source selection.*

Proof Under mandatory-only preprocessing, every admissible source sublibrary is the forced set together with exactly one of the 2^n residual subsets. Enumeration evaluates all of them. Its feasibility test is therefore true for precisely the feasible residual subsets, and its exact minimum is the residual optimum. Adding the selection-independent burden W_0 preserves objective order, and lifting is a feasibility- and burden-preserving bijection.

If the input already carries stronger preprocessing, the preprocessing propositions establish optimum-value preservation and a valid canonical lift; the solver does not reassert a bijection for eliminated columns. When the map also declares all optimizer identities reconstructable, its equal-coverage classes enumerate the additional optimal identities. Finally, the independent original-library checks exclude a corrupted lift or incidence-only certificate. The tagged-cover equivalence theorem identifies every checked lift with a safe source sublibrary. $\square$

S5.1.2 Requirement-mask dynamic program

Encode a requirement subset by an r -bit mask; union is bitwise OR. The implemented forward recurrence is the following.

1. Set $D_0[0] = 0$ and leave every other mask unreachable.
2. For $i = 1, \dots, n$ and every mask q , carry the skip transition $(q, D_{i-1}[q])$ and the take transition $(q \text{ OR } R_i, D_{i-1}[q] + w_i)$ into layer i , keeping the lower exact burden.
3. At equal burden, keep the first predecessor under the declared stable traversal, or retain every tied predecessor when complete-tie mode is requested.
4. Return the entry for the all-ones mask, add W_0 , reconstruct the original selection, and run the independent source-library certificate.

The implementation scans masks in increasing integer order and strategies in canonical identifier order, emitting the skip transition before the take transition. In representative mode it keeps the first exact predecessor; in complete-tie mode it stores the complete optimal predecessor DAG and expands it only during reconstruction. Thus iteration and representative tie selection are deterministic; hash-table iteration order does not determine output.

Theorem S5.2 (Exact requirement-mask dynamic programming). *For a valid finite tagged identity-closure instance, the recurrence above returns a minimum-burden safe source sublibrary and reconstructs at least one optimizer. In representative mode it uses $O(n2^r)$ exact-arithmetic and mask operations and $O(2^r)$ cost states plus $O(n2^r)$ reconstruction-aware predecessor records. If only optimum value were required, the two cost layers would use $O(2^r)$ memory. Complete-tie retention is output-sensitive: materializing T tied reconstruction vectors adds $O(Tn)$ bits and can be exponential in n .*

Consequently, the representative algorithm is fixed-parameter tractable in r : its bit-complexity is 2^r times a polynomial in the explicit input length. It is not polynomial in the full input size when r is unrestricted.

Proof For $i \in \{0, \dots, n\}$ and mask q , let $D_i[q]$ be the minimum exact burden of a subset of the first i residual strategies whose coverage union is exactly q , with $D_i[q] = +\infty$ if no such subset exists. We prove the stored invariant by induction on i . At $i = 0$, only the empty subset exists, so $D_0[0] = 0$ and all other states are unreachable.

Assume the invariant at $i - 1$. Every subset of the first i strategies either omits strategy i , in which case it supplies the skip candidate already characterized by D_{i-1} , or contains strategy i . Removing i from the latter subset leaves some predecessor mask q with optimal predecessor cost $D_{i-1}[q]$; reinserting it produces mask $q \text{ OR } R_i$ and cost $D_{i-1}[q] + w_i$. Conversely, every skip or take transition constructs a subset of the first i strategies with the stated mask and burden. Taking the minimum over these exhaustive and disjoint cases proves the invariant for

layer i . Stored predecessor records follow the same transitions, so every finite state has a valid path to the base layer. Equal-cost merging preserves the first predecessor under the declared deterministic traversal; retaining every equal-cost predecessor preserves the complete optimal predecessor DAG. Expanding that DAG preserves every tied residual identity, subject to the explicitly enforced output cap.

At $i = n$, a residual selection is feasible exactly when its union is the all-ones mask. Hence $D_n[2^r - 1]$ is the residual optimum. Adding the fixed mandatory offset and lifting preserve objective value directly under mandatory-only preprocessing; for a supplied stronger map, those properties are the content of the preprocessing propositions. Applying the tagged-cover equivalence proves safe-compression optimality. The independent exact certificate rechecks the source frontier, source closure, mandatory strategies, and burden rather than using the DP mask as its sole evidence.

There are 2^r masks and two relaxations per strategy-mask pair, giving $O(n2^r)$ recurrence operations in the unit-cost mask/arithmetic model. Two layers contain at most 2^r exact costs. The implementation additionally stores optimal predecessor records across the n layers, giving the stated reconstruction-aware bound. Exact rational sums and comparisons have bit cost polynomial in the explicit input bit length: sums of at most n encoded weights have polynomially bounded numerator and denominator length. Thus the bit-time bound is $2^r \text{poly}(|I|)$, which is fixed-parameter tractable in r and remains exponential when r is part of unrestricted input. $\square$

S5.1.3 Evidence boundary and counters

Both methods are exact finite computations, not Lean kernel verification and not mixed-integer solver evidence. The supporting mask-union declaration does not change that status. The standard result object records the exact objective and checked selections separately from wall-clock measurements. For enumeration, candidate count is the number of binary subsets tested. For the DP, a visited state is a reachable mask at one strategy layer and a transition is one evaluated skip or take relaxation; the final reachable-state count is reported separately. Runtime fields are single-run measurements split into preprocessing, search, and reconstruction/certification scopes. They are instrumentation, not an algorithmic scaling finding; scaling evidence requires the separately locked benchmark design.

S5.2 Approximation-guaranteed greedy construction

The identity-closure reduction also permits a fast constructive method with a classical performance guarantee. Fix all mandatory strategies, including the inactive strategy, and remove the tagged requirements they cover. Let U^0 be the residual requirement universe, let $W_0 \geq 0$ be the fixed mandatory burden, and let each residual strategy s have positive exact weight w_s and residual coverage $R_s \subseteq U^0$. For nonempty U^0 , write

$$d = \max_s |R_s|, \quad H(d) = \sum_{i=1}^d \frac{1}{i}.$$

The weighted rule is the standard set-cover greedy algorithm of [Chvátal \(1979\)](#), applied only after the semantic reduction:

1. initialize the selected set with every mandatory strategy and set $U \leftarrow U^0$;
2. while $U \neq \emptyset$, choose a residual strategy minimizing

$$\frac{w_s}{|R_s \cap U|}$$

among strategies with nonzero new coverage, breaking an exact tie by the lowest original strategy index in canonical identifier order;

3. add the chosen strategy and replace $U \leftarrow U \setminus R_s$;
4. optionally examine the chosen nonmandatory strategies in reverse selection order and delete one exactly when complete tagged coverage remains.

All scores and comparisons in the implementation use `Rational{BigInt}`. A separate cardinality variant replaces w_s in the score by one, equivalently choosing the largest number of newly covered requirements. That variant has the burden guarantee below only when all residual optional weights are equal; with unequal weights, no approximation guarantee for the actual retention burden is asserted.

Theorem S5.3 (Transferred weighted-greedy guarantee). *Suppose the source library is finite, closure is identity closure, every residual tagged requirement has a carrier, mandatory weights are nonnegative, and every nonmandatory weight is positive. If $U^0 \neq \emptyset$, the weighted greedy construction returns an innovation-safe source sublibrary of burden W_G satisfying*

$$W_G \leq H(d) \text{OPT},$$

where OPT is the minimum total burden of any safe source sublibrary and $d = \max_s |R_s|$. Deterministic reverse deletion returns burden $W_{GR} \leq W_G$ and hence satisfies the same bound. If U^0 is empty, the mandatory library is optimal and the factor is one.

Proof First suppose U^0 is nonempty. The residual instance is an ordinary positive-cost weighted set-cover instance. Chvátal's theorem applies to the rule minimizing cost per newly covered element and gives

$$G \leq H(d) \text{OPT}_R,$$

where G and OPT_R are, respectively, the greedy and optimal residual costs. The theorem permits any choice among exact minimum-ratio ties, so the canonical deterministic tie rule does not change the bound.

Every admissible source selection contains the mandatory strategies, and mandatory-only preprocessing removes no optional strategy. Thus $W_G = W_0 + G$ and $\text{OPT} = W_0 + \text{OPT}_R$. Since $W_0 \geq 0$ and $H(d) \geq 1$,

$$W_G = W_0 + G \leq W_0 + H(d) \text{OPT}_R \leq H(d)(W_0 + \text{OPT}_R) = H(d) \text{OPT}.$$

The tagged-cover equivalence then converts residual complete coverage plus mandatory retention into exact preservation of the source operating frontier and source identity closure.

Reverse deletion accepts a removal only after complete tagged coverage is rechecked. It never removes a mandatory strategy, and every removed optional strategy has positive weight. Therefore each accepted deletion strictly lowers burden, every rejected deletion leaves burden unchanged, and $W_{GR} \leq W_G$. Moreover, one reverse pass is enough for inclusion-wise

irreducibility: if a strategy cannot be removed at its examination, later deletions cannot restore any missing coverage. The same $H(d)$ bound follows immediately. If U^0 is empty, mandatory retention is already feasible and every optional positive-weight strategy is unnecessary, so the mandatory library is the exact optimum. $\square$

A primary-source near-tight fixture.

Chvátal’s example has m requirements, singleton set $\{j\}$ of cost $1/j$ for each $j = 1, \dots, m$, and a set covering the entire universe with cost $1 + \epsilon$, where $\epsilon > 0$ and $1 + \epsilon < H(m)$. Greedy selects the singletons in order $m, m - 1, \dots, 1$: when q requirements remain, singleton q has score $1/q$, whereas the universe set has score $(1 + \epsilon)/q > 1/q$. Its burden is $H(m)$, while the universe set alone is optimal with burden $1 + \epsilon$. The ratio $H(m)/(1 + \epsilon)$ tends to $H(m)$ as $\epsilon \downarrow 0$. The journal fixture realizes each requirement as an identity-closure module tag and lets the mandatory zero-weight inactive strategy cover the sole zero-frontier tag. Reverse deletion removes no singleton from the greedy endpoint.

Example S5.4 (Shared three-row greedy arithmetic). For $m = 3$ and $\epsilon = 1/12$, the singleton burden is $1 + 1/2 + 1/3 = 11/6$, the bundle burden is $1 + 1/12 = 13/12$, and the exact ratio is $22/13$. The Julia–Lean shared fixture checks only this arithmetic; it does not formalize Chvátal’s theorem or the transfer to safe compression.

Limitations and evidence boundary.

The guarantee is inherited set-cover theory; the contribution here is its proved transfer through the source-relative frontier-closure representation and exact postchecked implementation. The theorem does not apply to arbitrary nonidentity closure, because independent module rows need not represent closure complementarities. It does not give the backward heaviest-safe-first deletion heuristic an approximation guarantee, and it does not make cardinality greedy a weighted-burden approximation when weights differ. The returned certificate is exact finite computation: it rechecks mandatory retention, tagged coverage, the original frontier, the original identity closure, and burden. It is neither exhaustive optimality evidence, mixed-integer solver evidence, nor Lean kernel verification.

S5.3 Certified rechecked-deletion portfolio

Forward greedy constructs a cover from the mandatory library. Rechecked deletion instead starts from the source and removes one entry at a time. The common implementation performs a complete fresh scan after every accepted deletion. For each retained nonmandatory strategy, it constructs the trial library and recomputes exact source-frontier equality and exact source-closure equality from the original profiles and module memberships. A deletion is accepted only after both equalities, mandatory retention, tagged coverage, and burden reconciliation pass. All other certificates are then discarded.

The deterministic portfolio contains heaviest-safe-first, lightest-safe-first, maximum immediate burden release, minimum remaining unique-carrier exposure, and declared source order. Under additive burden, the release from deleting one strategy is exactly

its weight, so maximum immediate release and heaviest-safe-first are two names for the same ordering on this model; they are not treated as independent evidence.

For current retained set S , safe deletion candidate s , fixed tagged universe U_L , and tagged coverage R_t , define

$$E(s \mid S) = \left| \left\{ u \in U_L : \sum_{t \in S \setminus \{s\}} \mathbf{1}\{u \in R_t\} = 1 \right\} \right|.$$

The exposure rule minimizes $E(s \mid S)$, then maximizes exact immediate burden release, then uses the lowest canonical original strategy index. It uses only the source-defined incidence structure; it does not use dynamic value, held-out financial quality, realized return, or descendant outcomes. Heaviest and lightest rules break exact weight ties by lowest canonical original index. Declared-order deletion uses a complete caller-supplied permutation, or canonical original order by default.

The stochastic portfolio first draws a stable seeded permutation and then uses its earliest currently safe strategy at every fresh scan. Multi-start random deletion returns the lowest-burden certified endpoint across the declared starts; equal burdens are resolved by the lexicographically smallest selected-index tuple and then earliest start. The master seed, each derived seed, every random order, and every endpoint summary are retained.

Proposition S5.5 (Certified deletion endpoint). *Every algorithm in the rechecked-deletion portfolio returns a safe source sublibrary. Its reported endpoint is one-deletion irreducible after a complete exact final scan.*

Proof The source is safe relative to itself. At every accepted step, the trial library has been recomputed and certified to have the source frontier and source closure while retaining all mandatory entries. It is therefore safe, independently of the score used to choose it. Induction over the finite deletion trace gives safety of the endpoint. Every accepted deletion removes one retained nonmandatory strategy, so the trace terminates.

The final scan examines every retained nonmandatory strategy and records that its deletion fails exact source-relative feasibility. Hence no safe one-strategy deletion remains. Random ordering changes only the choice among already certified candidates and does not alter this argument. $\square$

The generic finite-trace safety implication in the first paragraph of the proof is Lean verified by `recheckedSafeDeletionEndpoint_feasible`. The declaration assumes a trace whose current-library redundancy certificates are already supplied. It does not verify the Julia scan, scoring, randomization, termination counters, or the reported final irreducibility scan; those clauses remain human proof plus exact Julia tests.

Safety and one-deletion irreducibility do not imply global resource efficiency. The singleton-bundle family in Theorem S6.1 remains a regression test: heaviest-safe-first and maximum immediate release return burden k while the global optimum has burden $1 + \epsilon$. Exact enumeration and mask dynamic programming, rather than any deletion status, supply small-instance optimum comparisons. Runtime counters are

instrumentation for the separately registered computational study, not scaling evidence in this draft.

S6 Worst-Case Local–Global Distinction

This section proves the parameterized counterexample used to separate exact safety certification from optimization quality. It concerns the named heaviest-safe-first rule, not every deletion order.

S6.1 Worst-case gap for heaviest-safe-first deletion

Define HSF to recheck exact source-relative safety at the current library, delete a safely deletable active strategy of greatest weight, and repeat until none remains. Ties may be resolved by any fixed rule; the construction below has a unique greatest-weight choice at its decisive step. For a minimization instance with positive optimum, its output ratio is the retained burden divided by the global minimum safe burden.

Theorem S6.1 (Unbounded gap for heaviest-safe-first deletion). *Let $k \geq 2$ be an integer and let $\epsilon \in \mathbb{Q}$ satisfy $0 < \epsilon < k - 1$. Consider one belief, identity module closure, modules $m_1, \dots, m_k$, and a mandatory inactive strategy of zero weight, zero profile, and no modules. For each i , let s_i have zero profile, weight one, and module set $\{m_i\}$. Let the bundle s_B have zero profile, weight $1 + \epsilon$, and all k modules. At the full source library:*

1. *every active strategy is safely deletable, and s_B is the unique heaviest safely deletable strategy;*
2. *HSF deletes s_B and stops at the library of all k singletons, which is inclusion-wise irreducible and has burden k ;*
3. *the bundle-only active library is the unique global minimum safe representation and has burden $1 + \epsilon$.*

Consequently, the output ratio of HSF is

$$\frac{k}{1 + \epsilon}.$$

The rule HSF therefore has no constant approximation guarantee over finite identity-closure innovation-safe compression instances. Moreover, one-deletion irreducibility alone has no constant worst-case burden guarantee relative to the global safe optimum.

Proof All active profiles and the mandatory inactive profile are zero. The source frontier is therefore zero and remains zero in every admissible sublibrary. Under identity closure, exact safety in this family reduces to covering all of $m_1, \dots, m_k$. A retained active set covers those modules if and only if it contains s_B , or it contains every singleton $s_1, \dots, s_k$.

At the full source, removing any singleton leaves s_B , so every singleton is safely deletable. Removing s_B leaves all singletons, so the bundle is safely deletable as well. Because $\epsilon > 0$, the bundle weight $1 + \epsilon$ is strictly greater than each singleton weight one. Thus it is the unique heaviest currently safe strategy, and HSF deletes it first.

After that deletion, singleton s_i is the only retained carrier of m_i . Deleting it would change the closure, so no further deletion is safe. More strongly, every proper sublibrary of the singleton endpoint omits some s_i and hence loses m_i ; the endpoint is inclusion-wise irreducible. Additivity of burden gives endpoint burden k .

The bundle-only active library is safe and has burden $1 + \epsilon$. Any safe library without the bundle must retain all k singletons and has burden $k > 1 + \epsilon$. Any safe library with the bundle and at least one singleton has burden strictly above $1 + \epsilon$, since active weights are positive. The bundle-only library is therefore the unique global optimum, and division by its positive burden gives the stated ratio.

Finally, fix the rational value $\epsilon = 1/2$. It satisfies the family condition for every $k \geq 2$, while the ratio is $2k/3$, which exceeds any proposed finite constant for a sufficiently large integer k . This proves the two negative guarantees: the named rule reaches these endpoints, and the endpoints themselves are one-deletion irreducible. It does not say that every deletion rule reaches them. $\square$

Example S6.2 (Shared finite gap arithmetic). At $k = 4$ and $\epsilon = 1/2$, the heaviest-safe-first endpoint has burden 4, the bundle optimum has burden $3/2$, and their exact ratio is $8/3$. The common Julia–Lean fixture checks this displayed arithmetic and the Julia generator separately checks the corresponding finite safety trace. The Lean declaration does not formalize the universal safety characterization, the named algorithm, or the unboundedness argument.

The deletion-order boundary is exact. If a rechecked maximal deletion sequence removes the bundle first, each remaining singleton is essential and the sequence stops at burden k . If it removes any singleton first, the bundle becomes essential for that singleton’s module; every remaining singleton stays redundant while the bundle is retained, so maximal deletion ends at the bundle-only optimum. Thus the theorem applies to the specified heaviest-safe-first rule and supplies bad one-deletion-irreducible endpoints, but it does not apply to every deletion order. In particular, it is not a lower bound for forward weighted-cover construction or for all certified deletion heuristics.

S7 Benchmark design and complete outcome accounting

The registered identity-closure benchmark contains 173 instances: 24 small exact, 128 structured, and 21 adversarial. The small family supports independent exact comparisons. The structured family varies source size, tagged requirements, incidence density, module overlap, bundles, unique carriers, weight dispersion, and frontier–module association. The adversarial family exercises the proved deletion gap, ties, duplicates, dominance, rare carriers, symmetry, and bundles. The instance registry fixes the realized comparison set; a requested but infeasible generator cell remains an unavailable instance.

S7.1 Methods and evidence

The method grid includes enumeration, mask dynamic programming, two registered HiGHS formulations, weighted and cardinality greedy, greedy followed by reverse

deletion, deterministic deletion orders, seeded random deletion, and 32-start deletion. Maximum immediate release and heaviest-safe-first are identical under additive weights and constitute one scientific method. The record keeps the two registered labels so that run-unit counts remain reproducible.

The independent audit checks original-coordinate mandatory retention, tagged coverage, complete frontier and closure equality, and exact burden. It accepts 3526 libraries among 3907 terminal run units. All 381 unsuccessful units remain in the denominator. Of 38 exact-capable instances, 31 have all required exact-method burdens available and equal, and 7 have incomplete comparisons. Equal burdens with different optimizer identities occur in 9 comparisons and are consistent with ties. Medium structured references are exactly rechecked HiGHS incumbents; solver-reported global optimality is distinct from the exact feasibility audit.

Table S1 All registered terminal run units. Solved categories sum to 3,526; every other category remains in the denominator of 3,907.

Terminal status	Run units
GENERATION_FAILED	352
IMPLEMENTATION_ERROR	6
MEMORY_LIMIT	14
NO_PRIMAL_CANDIDATE	1
PROCESS_INTERRUPTED	7
REJECTED_NONBINARY_SOLVER_CANDIDATE	1
SOLVED	3420
SOLVED_BY_EXACT_PREPROCESSING	106
Total	3907

The six implementation-error statuses arise when enumeration exceeds its registered applicability cap. They supply no enumeration result. Generation failures cover 16 source cells and remove complete requested structural combinations. No failed instance is replaced. No terminal run is a time-limit outcome; therefore no timeout-gap distribution is estimated.

S7.2 Finite deletion-gap check

The following registered fixtures compare the observed burden ratio with the proved singleton–bundle family. They check the finite implementation, while the theorem establishes the unbounded family result.

S7.3 Interpretation and execution conditions

Full preprocessing commonly resolves structured instances completely. This supports the importance of forced carriers in this design but leaves limited evidence about difficult residual search. Multi-start completes all starts on 143 instances and improves on its first start on 96; successful sensitivity runs do not replace failed primary runs.

Scale	Parameter	Reference evidence	Observed ratio	Theoretical ratio
small	$k = 4, \epsilon = \frac{1}{2}$	exact finite	$\frac{8}{3}$	$\frac{8}{3}$
medium	$k = 16, \epsilon = \frac{1}{2}$	HiGHS-OPTIMAL, rechecked	$\frac{32}{3}$	$\frac{32}{3}$
large	$k = 64, \epsilon = \frac{1}{2}$	HiGHS-OPTIMAL, rechecked	$\frac{128}{3}$	$\frac{128}{3}$

Table S2 Registered adversarial instances for heaviest-safe-first. Equality between observed and theoretical ratios is a synthetic fixture check of the human-proved family, not an empirical approximation guarantee.

Execution used Julia 1.12.6 and HiGHS v1.15.1 on an Apple M1 Pro, with 8 concurrent one-thread processes. Times include compilation and worker contention, and the execution record identifies the source worktree. They are not steady-state speed estimates. The registered performance profile uses unlike success targets across methods. The registered HC3 models also omit within-instance dependence and cannot recover missing generator cells. Neither output supports the article’s comparative or inferential claims. The published comparison instead uses exact agreement, certified burden, preprocessing, and the complete status census.

The data files named in the reproduction manifest supply the complete instance registry, exact-agreement records, method-level burden summaries, preprocessing and multi-start summaries, and terminal records. The PDF reports aggregates; no source row is removed from those current study records.

S8 Financial and mechanism methods

The point-in-time portfolio study is retrospective development evidence. Its historical setting was known during design. Its sequential freezes protect the stated within-study information boundary, without making the historical sample an independent confirmatory test.

S8.1 Financial universe, strategy grammar, and timing

For origin year y , years $y - 5$ through $y - 3$ form the source library; years $y - 2$ through y provide compression profiles; year $y + 1$ supplies proposal estimates and freezes policy choices; year $y + 2$ evaluates those choices. No evaluation value enters any choice. The primary universe contains the 100 most liquid eligible US common equities at the origin. The separate ETF replication selects up to 50 eligible plain ETFs and requires at least 20. One ETF origin fails its universe gate and remains unavailable. Eligibility uses predecision data and does not require subsequent survival.

The 96-specification grammar crosses three signals, two filters, two holding horizons, two allocation rules, two gross exposures, and two exit rules. Signals use 20- or 60-session momentum or five-session mean reversion; filters are unrestricted entry or a 100-session trend filter. Allocation is equal weight or inverse trailing volatility. Exposure is subject to the frozen security cap and a 10% annual portfolio-volatility ceiling; unused exposure remains cash. Intended weights formed at close t first earn the return indexed $t + 2$. Observed delisting returns are earned by the effective position and

followed by cash. A required missing return makes the path unavailable; it is never imputed as zero. The current grammar and burden registries fix the remaining rule values and identifiers.

Formation and compression each use 12 profile rows: three years crossed with 5- and 30-basis-point one-way costs and risk aversion 1 and 3. Each floating profile is rounded to the nearest 10^{-12} and represented as an exact rational for comparison. Thus frontier equality is exact for the recorded table, without treating measured returns as exact economic quantities. The source contains all formation-frontier carriers plus the first three formation-ranked carriers of each required capability. It is frozen before compression profiles are computed. Each cell has 31 declared capabilities, including atomic rules, interfaces, relevant point-in-time data, corporate actions, aggregation, and cost/capacity support.

S8.2 Burden, compression, and proposal sets

A strategy's burden is the exact integer sum of its components in the frozen burden registry. It is a validation-computation index, with zero burden for mandatory cash. The safe arm preserves the source frontier and full capability union. The comparator first solves frontier-only retention and then adds whole source entries in descending mean compression score, breaking ties by identifier, whenever they fit within the safe arm's realized burden cap. Every endpoint is independently checked in the unprocessed coordinates. 3 safe cells are resolved by exact preprocessing; 34 receive HiGHS OPTIMAL status and an exact feasibility/burden postcheck.

Proposal actions comprise the grammar candidates excluded from the source docket and enabled by the retained requirements. The comparator action set is nested in the safe set before reading proposal scores. The complete candidate ledger retains source-docket exclusions, missing capabilities, rejected candidates, cash, and unavailable cells. The analysis unit is the origin; candidate specifications and daily returns are not independent replications.

S8.3 Certainty equivalent and protected selection

For gross daily portfolio return r_t , one-way turnover q_t , and net return $n_t = r_t - 0.0005q_t$, the primary annual certainty equivalent is

$$\text{CE} = 252 \bar{r} - 252 \overline{0.0005q} - \frac{1}{2} 3 252 \widehat{\text{Var}}(n),$$

with the registered sample-variance convention. The mean, cost, and risk penalty are reported separately. The economic adoption hurdle is 0.0025 in annual CE units.

Proposal uncertainty uses 5,000 paired circular moving-block bootstrap draws with 20-session blocks. A draw applies the same resampled session indices to each candidate and baseline. If δ_i is an estimated annual CE difference and se_i is the standard deviation of its bootstrap differences, the reference maximum for each draw uses $(\delta_i - \delta_i^*)/\text{se}_i$, with zero used when the standard error is zero. The 90th percentile of that maximum is the one-sided critical value c , giving lower bound $\delta_i - c \text{se}_i$.

The comparator selects cash unless an eligible candidate's simultaneous lower bound over cash exceeds the hurdle. The safe policy defaults to that exact choice. Because

the comparator is selected from proposal data, the safe reference family includes every complete safe-candidate by complete comparator-candidate contrast, including cash. Replacement requires the corresponding simultaneous lower bound to exceed the hurdle. Family shrinkage ranks only candidates that pass this gate: its center is the family mean, its latent variance is the nonnegative part of between-path sample variance minus median squared standard error, and its reliability weights shrink each candidate toward that center. Identifier order resolves remaining ties.

S8.4 Origin-level outcomes and robustness

All 19 common-equity pairs choose the same action, so their held-out contrasts are exactly zero. One of the 18 complete ETF pairs replaces the comparator and realizes a negative contrast. The complete origin-level record follows.

Table S3: Origin-level point-in-time held-out contrasts. CE differences are innovation-safe minus frontier-only. A dash marks the prespecified failed ETF universe gate; it is retained, not imputed.

Origin	Equity CE diff.	Same choice	ETF CE diff.	Same choice
2005	0.0000	yes	—	—
2006	0.0000	yes	0.0000	yes
2007	0.0000	yes	0.0000	yes
2008	0.0000	yes	0.0000	yes
2009	0.0000	yes	0.0000	yes
2010	0.0000	yes	0.0000	yes
2011	0.0000	yes	0.0000	yes
2012	0.0000	yes	0.0000	yes
2013	0.0000	yes	0.0000	yes
2014	0.0000	yes	0.0000	yes
2015	0.0000	yes	0.0000	yes
2016	0.0000	yes	-0.0390	no
2017	0.0000	yes	0.0000	yes
2018	0.0000	yes	0.0000	yes
2019	0.0000	yes	0.0000	yes
2020	0.0000	yes	0.0000	yes
2021	0.0000	yes	0.0000	yes
2022	0.0000	yes	0.0000	yes
2023	0.0000	yes	0.0000	yes

The prespecified cost, risk-aversion, and hurdle grid preserves common-equity choice equality. ETF differences remain at the same origin and disappear at higher costs. Leave-one-origin-out summaries aggregate frozen choices without reselection. Unregistered larger-universe and closure-sham variants remain unavailable. Forced-maximum and full-family search summaries are diagnostics and never feed back into selection. Capacity calculations score frozen choices and retain missing inputs, incomplete paths, and participation-cap breaches.

S8.5 Calibrated synthetic bridge experiment

The two libraries have equal current frontier and burden; only the safe one can run the specified descendant project. The parameters are $\beta = 0.99$, $\rho = 0.98$, $\pi = 0.90$, and $d = 1$, so success probability is 0.882. The net bridge margin is $M = \beta^d \rho^d \pi G - \kappa$. An oracle researches precisely when $M > 0$; the learner researches when its noisy estimate exceeds an independently calibrated threshold.

Calibration uses 2479 complete financial proposal paths and 623647 residual observations, after removing cross-strategy date means within cells. It estimates scale, lag-one dependence, and tail thickness. Simulated proposal noise follows a stationary AR(1) with variance-normalized Student innovations, 100 burn-in sessions, 756 proposal sessions, and annualization 252. A separate 4,096-world null simulation fixes the threshold at the greater of the 0.0025 hurdle and its 95th percentile, giving 0.027008117603021155. Research cost and powered margin are each twice that threshold.

The four margins are negative one threshold, zero, 0.0025, and positive two thresholds. Each has 4,096 test worlds. Common random numbers pair regimes within a world; calibration and test seeds are disjoint. Mean intervals use mean plus or minus 1.96 standard errors; adoption uses Wilson intervals. All four regimes and all worlds remain in the ledger. The primary powered-oracle gate requires a positive lower endpoint and inclusion of the analytic margin. The powered learner must have a positive lower endpoint and at least 90% adoption. Null and adverse point adoption must not exceed 6.5% and 1%, respectively.

The article reports the full four-regime table. Its null point adoption is 5.76%, while its Wilson upper endpoint is 6.52%. The low positive dose does not yield a positive learned interval; the powered one does. The powered margin is a designed dose, and the experiment supports its declared synthetic mechanism rather than a market premium. Structural assertions check frontier, burden, and bridge availability in every world.

S8.6 Input access

Public financial outputs contain aggregate results, structural counts, and cryptographic seals. Raw CRSP/WRDS rows, security identifiers, return paths, and selected identities derived from licensed returns are excluded. Full source-data replication requires independent licensed access. Public aggregate tables and the synthetic ledger have separate reproducible checks identified in the reproduction guide.

S9 Theorem and Evidence Correspondence

The table records the proof and verification scope of the results used in this paper. A supporting finite fixture never upgrades the status of a universal result, and solver evidence establishes none of the mathematical claims in this table.

Table S4: Evidence status of the theory and algorithm results.

Result	Verification status	Supporting evidence and boundary
Tagged-cover equivalence and weighted objective identity	Lean verified	Human proof; exact Julia boundary fixtures. The Lean result covers the encoded universal biconditional and burden identity, not complexity or approximation.
Identity-closure	Human proof only	Exact reduction fixtures and Julia tests; no Lean complexity framework.
NP-completeness	Human proof only	Exact rational family fixtures and Julia tests; Lean checks one finite arithmetic case only.
Unbounded	Human proof only	Lean verifies selected finite feasibility cores; objective preservation, reconstruction, propagation, and fixed-point claims are exact fixtures and Julia tested, not formally verified.
heaviest-safe-first gap	Human proof only	Exact cross-method fixtures and Julia tests; Lean mask identities do not formalize the recurrence invariant or complexity bound.
Preprocessing propositions	Human proof only	The external Chvátal theorem is not re-formalized; finite ratios are exact fixtures and Julia tested.
Enumeration and requirement-mask DP correctness, complexity, and FPT status	Human proof only	Julia tests cover scoring, stale-certificate invalidation, reproducible randomness, counters, feasibility, and irreducibility. No approximation quality follows.
Weighted-greedy transfer and reverse deletion	Lean verified for an abstract supplied safe trace; human proof only for concrete-suite termination and final-scan irreducibility	
Rechecked-deletion feasibility		

The Lean axiom audits recorded for the tagged-cover and preprocessing declarations print only `propext`, `Classical.choice`, and `Quot.sound`. The shared Julia–Lean fixture prevents manual transcription of selected finite rational values. The reproduction manifest identifies the required formal sources under `formal/StrategyInnovation/Optimization/`, `formal/StrategyInnovation/Fixtures/`, and `formal/StrategyInnovation/Audit/`.

S10 Reproducing the paper and its evidence

The accompanying editable source package builds the article, this resource, and their combined PDF. Run `python3 build.py -output ./release` after installing Python with `pypdf`, a LaTeX distribution with `latexmk`, and Poppler. The builder checks references, embedded fonts, searchable text, page counts, and internal destinations in the combined file. Its source manifest and SHA-256 checksums identify every distributed input. The current package is available from the repository’s [versioned preprint release directory](#).

S10.1 Public evidence and source binding

The `evidence/` directory contains the complete benchmark terminal CSV, instance and seed registries, exact-agreement and method summaries; the financial study’s public aggregate evaluation manifest and burden/capability registries; and the calibrated synthetic experiment’s configuration, calibration, regime and seed registries, summary, and complete world ledger. The machine-readable `REPRODUCTION_MANIFEST.json` records original repository paths and SHA-256 hashes for these files and the selected scientific sources. Internal study identifiers locate the exact implemented experiments.

The repository source commit is `bb58f33fd42f82f51b2866b44a666c3094f2932a`. The document source archive is self-contained for PDF rebuilding and public evidence inspection. Full computational replication additionally requires the repository’s pinned Julia and Lean environments and the scientific source files identified in the manifest. File hashes, rather than a moving branch name, bind the computational inputs.

S10.2 Mathematical and algorithmic verification

The main article and Sections S1–S6 contain the human arguments. Selected Lean modules cover the projection, tagged-cover result, finite preprocessing cores, and supplied safe-deletion traces. The correspondence in Section S9 states which complete results remain human proofs. The reproduction guide gives focused Julia fixture checks and Lean module commands; a successful finite fixture is not a universal proof.

S10.3 Numerical reproduction

The benchmark’s saved terminal records retain unsuccessful and unavailable runs. The exact audit reads those outcomes and rechecks returned libraries; regenerating a benchmark is a separate computation whose machine timing may change. The financial tables can be checked against distributed public aggregates, but portfolio paths and policy selection require independent licensed data access under the specified staging contracts. The synthetic experiment can be rerun from its public calibration and fixed configuration; its calibration and test worlds are distinct. The public calibration values suffice for this rerun; recalibrating those values from source market returns requires the licensed paths.

The accompanying `README.md` provides commands for each of these tasks and distinguishes rebuilding documents, checking public records, and recomputing scientific outputs. No raw or row-level licensed observation is included in the distribution.

Generative-AI disclosure

OpenAI ChatGPT and Codex assisted with LaTeX drafting and editing, software review, and repository-based analysis and reproducibility tasks. The author remains solely responsible for the resource. No AI system is an author and no generative-AI image is used.

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